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SOUTHERN GEOSCIENCE CONSULTANTS

MEMORANDUM

TO	Kieren Whittock, Andy Border - Hastings Technology Metals Limited (ASX : HAS)
FROM	Jarrad Trunfull, Russell Mortimer - SGC
DATE	01/06/2018
REPORT NO.	**SGC3347**
RE	**Yangibana Project - Aeromagnetic Prospect/Target Modelling**

1 INTRODUCTION

Modelling of magnetic data has been performed over Hastings Technology Metals' (Hastings) Yangibana project by Southern Geoscience Consultants (SGC), which is prospective for Rare Earth Elements (REEs), particularly Neodymium and Praseodymium. The data used was from an aeromagnetic survey flown over the area in 2016, the details of which are given in **Table 1**. This data was processed by SGC in 2016.

Table 1. Details of aeromagnetic survey used for magnetic modelling.

SURVEY NAME	METHODS	JOB #	CONTRACTOR	SURVEY YEAR	FLIGHT LINE SPACING (metres)	MEAN TERRAIN CLEARANCE (metres)	FLIGHT LINE DIRECTION (degrees)	DATA STATUS
Yangibana	MAG RAD DEM	F16020	Thomson Aviation	2016	25	30	030/210	Confidential

Modelling was performed over 5 targets/areas, which are described in **Table 2** and shown in **Figure 1**.

Modelling has also been previously performed by SGC over the Auer North prospect. This area has since been drill tested and the modelling is reviewed in this memorandum.

Over the Target 5 area, 'depth filtering' (a series of upward continuation filters) has been performed in an attempt to reveal deeper structure/bodies of potential interest.

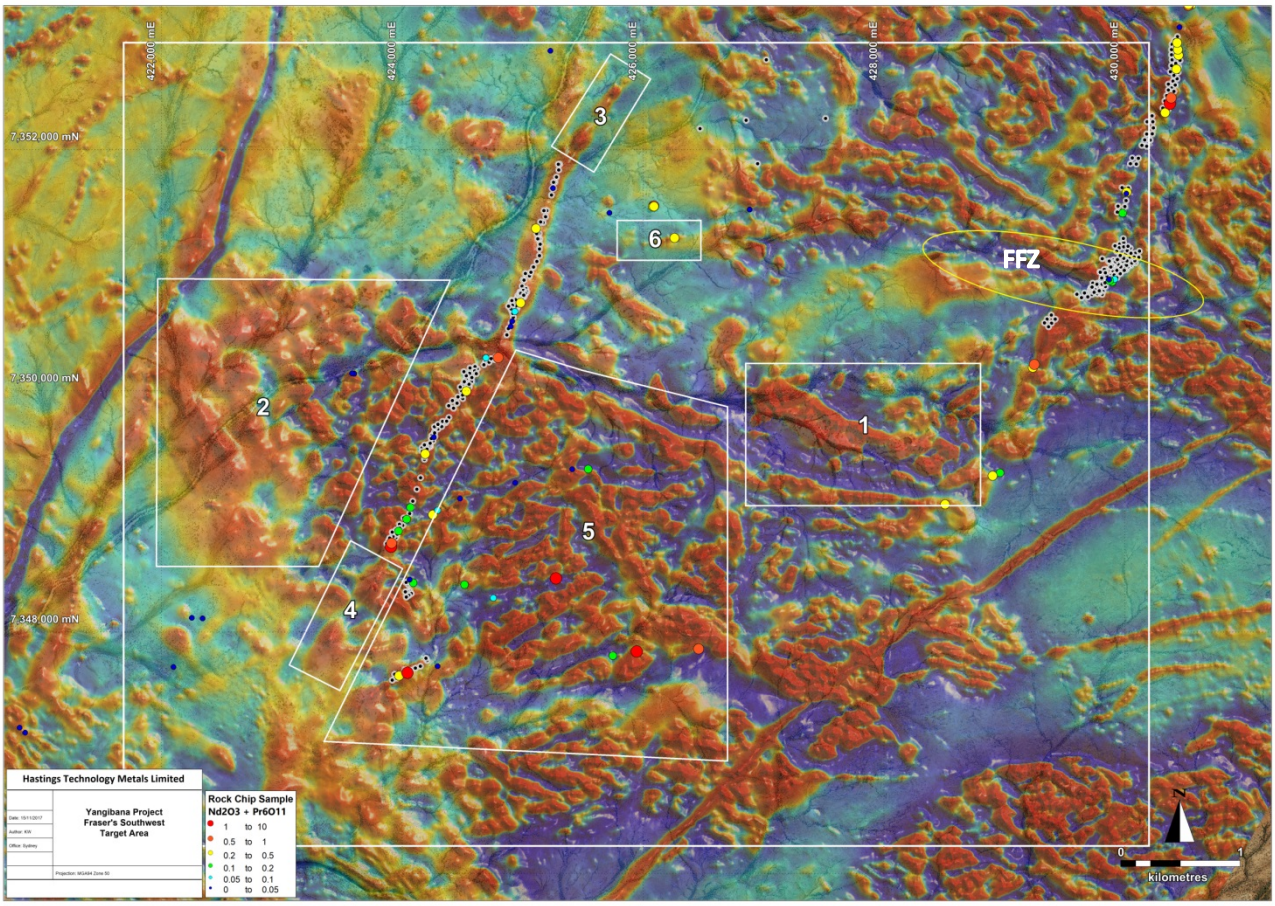


Figure 1. Target areas for modelling.

Table 2. Targets/Prospects for Modelling

Target	Description
Target 1	Large, strongly magnetic feature with interesting signature. No other information known about prospect.
Target 3	Possible northern extension of the Auer/Auer North linear magnetic feature.
Target 4	Possible southern extension of the Auer/Auer North linear magnetic feature.
Target 6	Small linear magnetic feature, some soil sampling indicative of possible interest.
Fraser Fault Zone	Large zone of mineralisation in heavily structurally deformed area.

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2 3D INVERSION MODELLING

2.1 Target 1

Three-dimensional unconstrained inversion modelling was performed over the Target 1 feature to derive information about its depth, size and strength. No other information has been collected over this feature (mapping, sampling, etc.) prior to this study, is believed to be under cover. An approximately 2.6 x 2.0 km area of TMI data around the feature was inverted using the UBC v5 code to produce a 3D smooth model of magnetic susceptibility (SI).

The resultant modelled body (**Figure 2**) is large, discrete and with a strong magnetic susceptibility. The shape implies a potential fold structure (synform or antiform) that has been 'cut' and offset multiple times by ENE trending brittle structures (**Figure 2**). The body appears to have a large depth extent.

Follow-up forward modelling was performed over Target 1 to complement the inversion model and further understand the structure of the feature (**Section 3.2**).

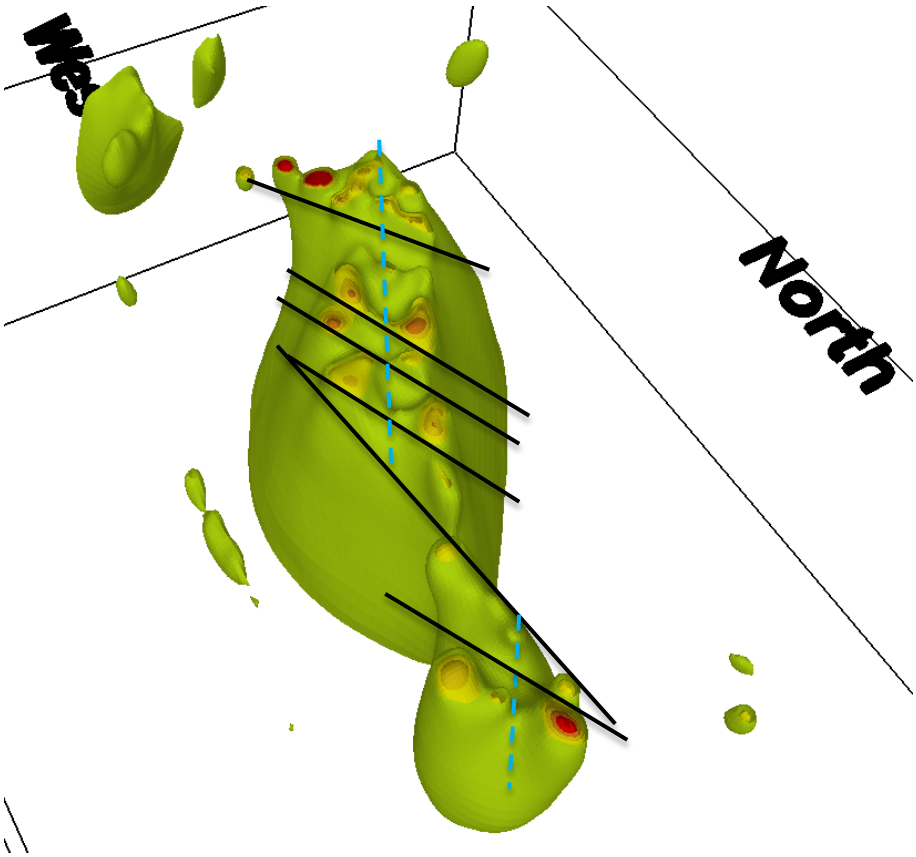


Figure 2. Image of magnetic inversion (SI ≥ 0.01) showing an ambiguous fold structure. Interpreted fold axes annotated with blue dashed lines and interpreted faults annotated with black lines.

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2.2 Fraser Fault Zone

The Fraser Fault Zone is a large, structurally complex zone that hosts significant REE mineralisation. The mineralisation here does not seem to be magnetic itself, however magnetic modelling has been undertaken to better understand the structure and geometry of the surrounding geology, and by correlation, of the mineralisation.

Two-dimensional modelling of the Fraser Fault Zone (see **Section 3.6**) was performed, but was considered difficult due to the complex structure, variable magnetic signature/Q-factor of units, and strong, variable regional field. As a result, a three-dimensional unconstrained inversion was performed to better understand the behaviour of the geology.

Figure 3 shows an image of inversion shells (< -0.002 SI) along with drilling (coloured by Nd_2O_3+Pr_6O_{11} %, Blue = 0%, Red = 1%) and an interpreted mineralisation horizon (red). The mineralisation here appears to be associated with low/non-magnetic bodies and structures that are running through the area.

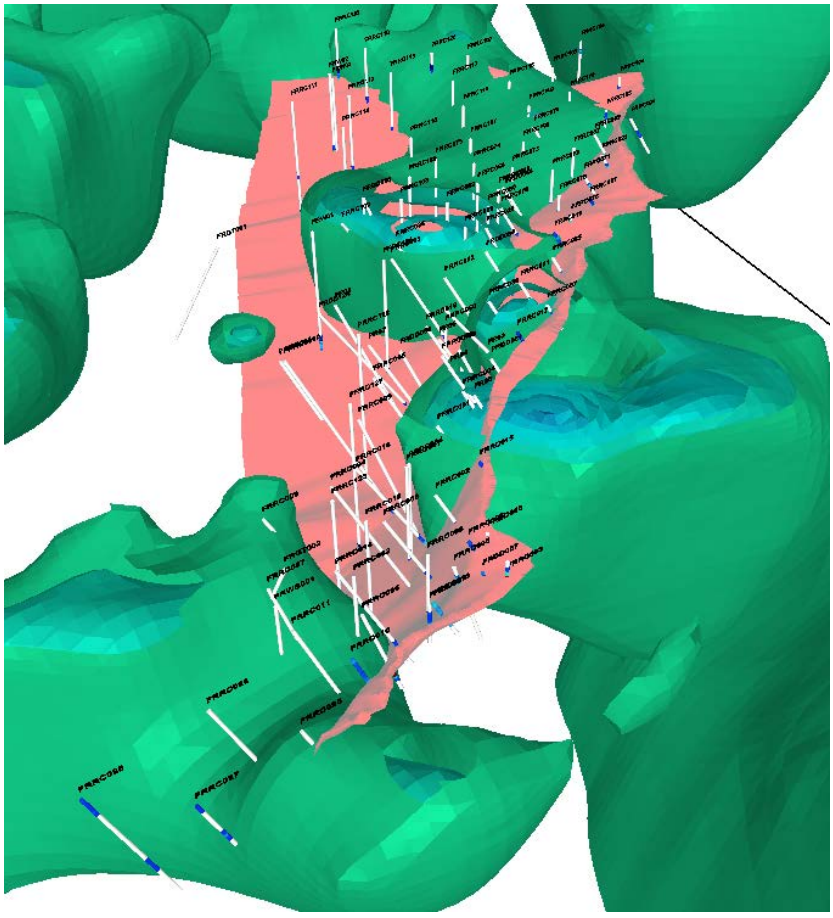


Figure 3. Inversion over Fraser Fault Zone prospect showing drilling, know mineralisation polygon (red) and magnetic shells (<-0.002 SI). The mineralisation in this case appears to be associated with the zone of low magnetism and/or a structure.

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3 FORWARD MODELLING

Forward modelling has been carried out over Targets 1, 3, 4, 6 and the Frazer Fault Zone using Potent software in an attempt to understand the shape, depth and width of bodies that may contain or be linked to REE mineralisation.

The modelling results are described below and shown in **Appendix A**.

3.1 Potent Forward Magnetic Modelling

The Potent modelling software calculates the theoretical 'induced magnetisation' from simple shapes (dykes, cylinders, blocks, spheres, etc.) which can be compared to the TMI profiles from field surveys. The orientation, dimensions, location and magnetic susceptibility of these model bodies can be adjusted until the calculated response matches the observed response; this process is referred to as 'forward modelling'. Limitations of the forward modelling routine include;

- There is no unique solution for modelling an observed TMI field, in most scenarios there are more variables than there are known parameters. There are trade-offs between different parameters that produce similar results. For example, a wider and less magnetic body may produce a similar response to a thinner and more magnetic unit. If some of the parameters can be constrained from known geological information (drill-hole data or magnetic susceptibility measurements) then the range of solutions can be reduced producing a higher degree of confidence in the modelling.
- The induced magnetic field decreases strongly with distance. This means that the magnetic material closest to the magnetometer has the strongest response and generally produces better constrained models than magnetic material that is further from the sensor. For ground or airborne TMI surveys, the top of the magnetic bodies are easier to model than the deeper extents. For borehole surveys, magnetic material closer to the drillhole is easier to model than the magnetic material far from the drill hole.
- Multiple magnetic bodies that are closer to each other than their distance to the magnetometer cannot be resolved as separate sources – their magnetic fields will superimpose to look like that of a single magnetic body. What has been modelled as a single magnetic body may actually be composed of a number of smaller and closely spaced bodies.
- When modelling magnetic bodies the thickness and susceptibility values cannot be resolved individually. The induced magnetic field will be a product of the volume of magnetic material (i.e. thickness x magnetic susceptibility) such that a 10m thick body with susceptibility of 1.0 SI will produce the same result as a 20m thick body of 0.5 SI.
- Potent forward modelling uses simple shapes to model what can be very complex geological scenarios. The best results are achieved where simple geology is modelled using TMI profiles that are close to being perpendicular to the dominant strike direction. Modelling TMI profiles that are oblique or parallel to strike can be very difficult, particularly where geology is very complex.
- The observed TMI response can be affected by remanent magnetisation, anisotropic magnetisation or demagnetisation which are difficult to quantify and account for. These effects distort the TMI anomaly (from the induced magnetisation) and will introduce errors into the modelling efforts.
- Forward modelling is often referred to as 2D modelling. The models generated apply only to the profile being modelled. Data from nearby lines can be used to help constrain the orientation (strike, dip, plunge, length) of the model bodies, in which case the modelling is sometimes referred to as 2.5 D modelling. The models generated for each profile are valid only for that data on that profile; the results should not be extrapolated too far away from the modelled data.

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In general the magnetic modelling can be improved by constraining parameters with known information and by recording magnetic profiles closer to the magnetic source rocks. For the modelling of airborne TMI data this means the modelling results can be improved by recording ground or borehole TMI profiles and using drilling data to constrain the dimensions and magnetic susceptibility of the magnetic model bodies.

An example of a 2D model section is shown below in **Figure 4**. For each model profile the TMI (top panel) and the two derivatives TMI_dz (first vertical derivative of TMI – second panel) and Manalytic (analytic signal of TMI – third panel) have been modelled. The derivatives help to isolate the response of closely spaced magnetic bodies and identify the edges of shallow targets. The observed profiles are shown in blue and the calculated profiles in red. The lowest panel on each profile shows the cross section of the profile with the relevant model bodies and drill holes projected onto the section. The AMAG survey flight path is shown as a green trace and the terrain as a brown trace (all RL positions are relative to the aircraft GPS heights). The model body cross sections are filled in and the ends are projected onto the section (near side is solid and the far side is dashed). The cross section direction is labelled in the lower right-hand corner ("AZ=89.9 deg")

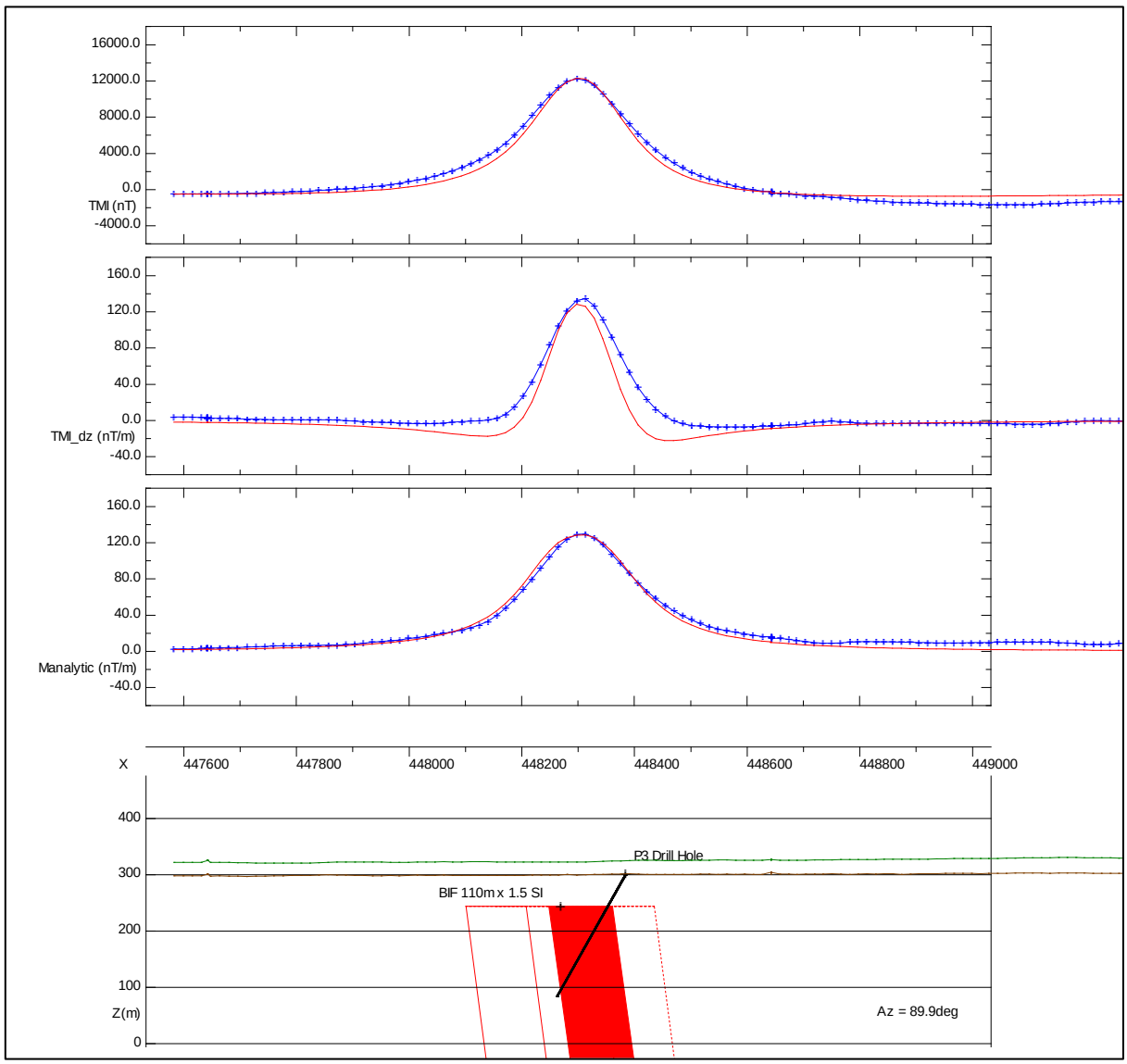


Figure 4. Example of a Potent 2D model profile.

3.2 Target 1

Forward modelling was carried out on three flight lines over the Target 1 feature to complement the 3D inversion – Lines 110370, 110430 and 110540. Three dyke-like bodies were used to model the feature. Modelling implies the body could represent a magnetic antiform with two or more folded magnetic horizons, with vertically to steeply dipping limbs.

3.3 Target 3

Target 3 comprises the (possibly mineralised) northern extension of the Auer magnetic trend. Survey lines in this area are angled parallel to the trend, with no perpendicular tie lines crossing directly over the features of interest. As a result, the data across multiple parallel survey lines (109210 to 109310) was extracted for modelling. This had the disadvantage that station spacing was limited to line spacing (25m), but was seen as better than attempting to model gridded data, which was smoothed and could not be modelled in conjunction with the parallel line data in Potent.

Modelling produced two relatively thin (15-20m), steeply dipping bodies with steep dips, although the dips are ambiguous given the low sample spacing.

3.4 Target 4

Target 4 is similar to Target 3 in that it is the possible southern extension of the Auer magnetic trend. Unlike Target 3, there were perpendicular tie lines crossing directly over the features of interest in this area. Lines 109360, 109400, 190170 and 190190 were utilised in the modelling.

Modelling produced two relatively steeply dipping bodies, 20-30 metres wide and plunging to the south. Surrounding and 'deeper' magnetic features were also modelled, including one body which was given a negative magnetic susceptibility in order to fit the regional signal.

3.5 Target 6

Target 6 is a thin magnetic anomaly coincident with an anomalous surface sample. Lines 109670 and 190340 have been used for modelling.

Magnetic modelling shows the body to be extremely thin, shallow and strongly magnetic – possibly with an element of remanence. 'Inversion' of the polygon within Potent produced an unrealistic model located above the surface. Review of the magnetic image leads belief that the body is possibly a shallow, magnetic palaeochannel feature. It is coincident with modern drainage when viewed in conjunction with satellite imagery.

3.6 Fraser Fault Zone

Two-dimensional modelling was initially performed at the Fraser Fault zone to determine depths, dips and geometries of units in the target area. Lines 110780, 110860, 110970 and 190430 were used for modelling.

Due to the strong regional/variation in basement magnetism, line-to-line modelling was difficult and where it was difficult to directly fit the TMI data, modelling focussed on the TMI 1VD data in order to determine the geometries of the units.

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4 AUER FORWARD MODELLING REVIEW

The Auer and Auer North forward models produced by SGC in 2016 have been compared against drilling results and the derived mineralisation polygon (**Figure 5**).

Drilling shows the width of the mineralisation to be thin (<3 metres), whereas the polygons of widths of 5 and 20 metres were modelled in order to fit this data. This could reflect that the magnetic halo around the mineralisation is wider than the mineralisation itself.

The mineralisation polygon supplied by Hastings shows the mineralisation to be reaching surface, and the dip to be steeper than the model. The magnetic models were located at approximately 50 to 80 metres depth – although the modelling could be better constrained now, it is likely the top layer of the mineralisation has been oxidised by weathering and has therefore lost magnetism (as magnetite oxidises to hematite). The dips of the magnetic model tend to reflect a best/average estimate which was not too far from reality in this instance.

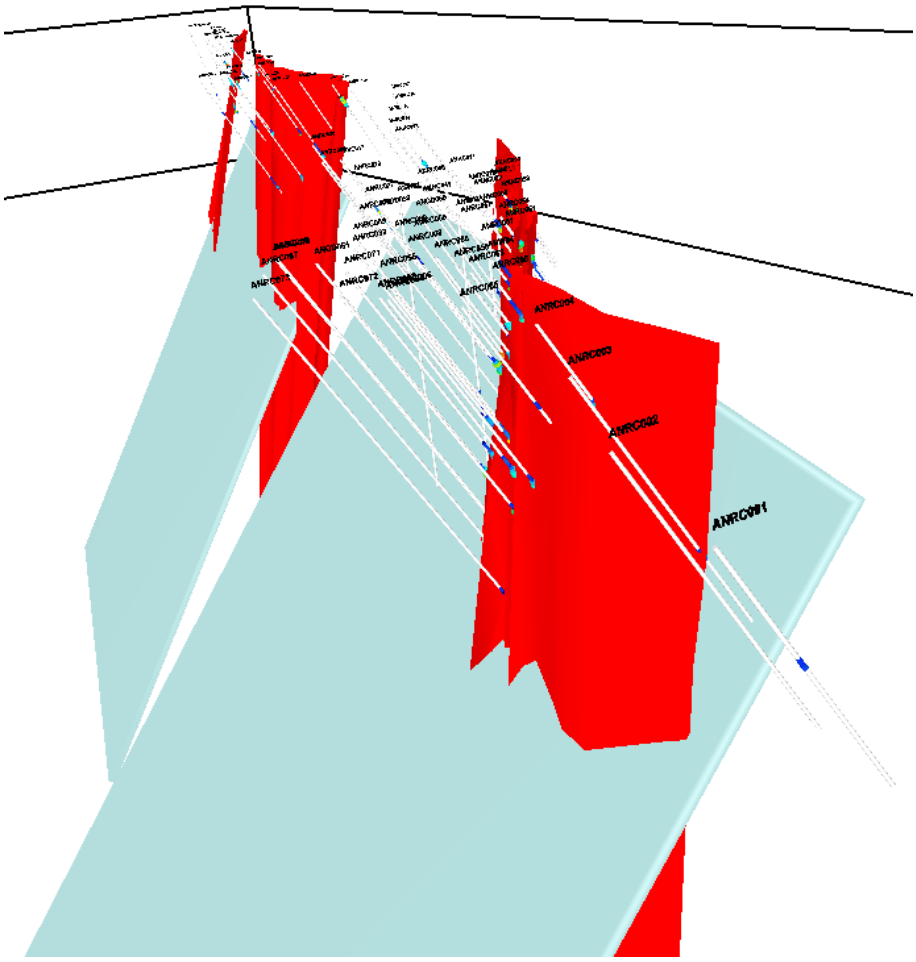


Figure 5. Comparison of 5m-wide magnetic models (blue), drilling and mineralisation polygon (red). Drilling coloured with Nd_2O_3+Pr_6O_{11} % (Blue = 0%, Red = 1%).

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5 DEPTH FILTERING

Filtering of the magnetic data was undertaken in order to suppress high frequency/noisy surface signal with the aim of revealing deeper components in the Target 5 area of interest (ref. **Figure 1**). A series of upward-continuation filters were applied to the RTP data, which simulate the data being collected further away from the source (e.g. a 20 metre upward-continuation filter simulates the data being collected from 20 metres higher; since the original data was collected at 30 metres terrain clearance, this would be a virtual terrain clearance of 50 metres).

Upward continuation filters of 5, 10, 20, 50 and 100 metres were performed on the RTP data, and a first vertical derivative was then applied to highlight structural features (**Figure 6**). With each increase in continuation, shorter wavelength features (e.g. shallow, high magnetic) are merged/suppressed and longer wavelength features are accentuated to highlight deeper and more major structures and geometries. This information can then be used to 'zone in' to areas of interest.

A number of candidate larger, deeper magnetic bodies have been highlighted by this depth filtering (50-100m upward continuation imagery) and should be assessed from a geological perspective as to exploration interest (**Figure 7**).

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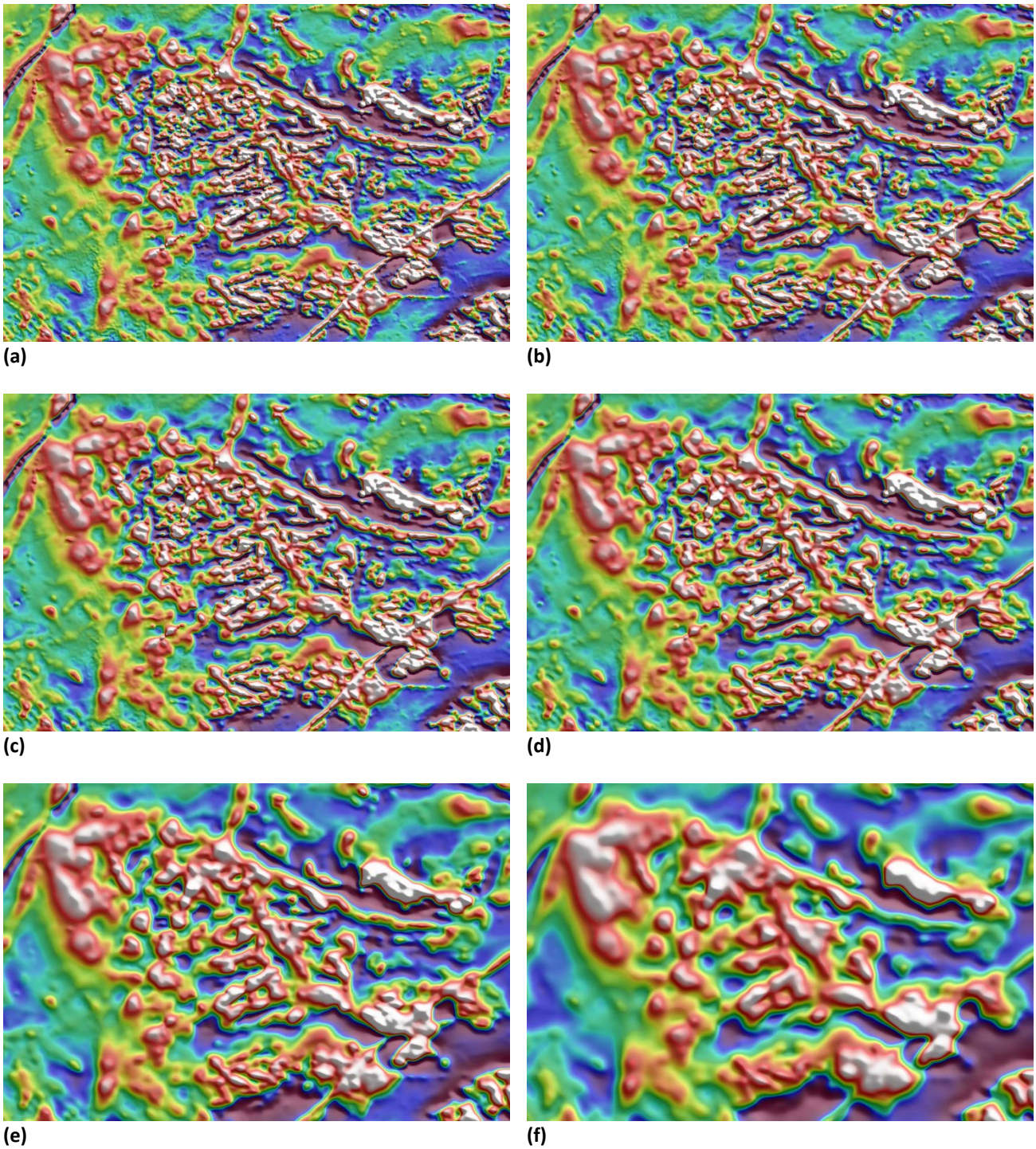


Figure 6. Depth filtering over Target 5 area (ref. Figure 1) using Upward Continuation (UC) of RTP magnetic data in order to accentuate deeper features – (a) RTP 1VD; (b) RTP UC 5m 1VD; (c) RTP UC 10m 1VD; (d) RTP UC 20m 1VD; (e) RTP UC 50m 1VD; (f) RTP UC 100m 1VD. Non-linear colour stretch.

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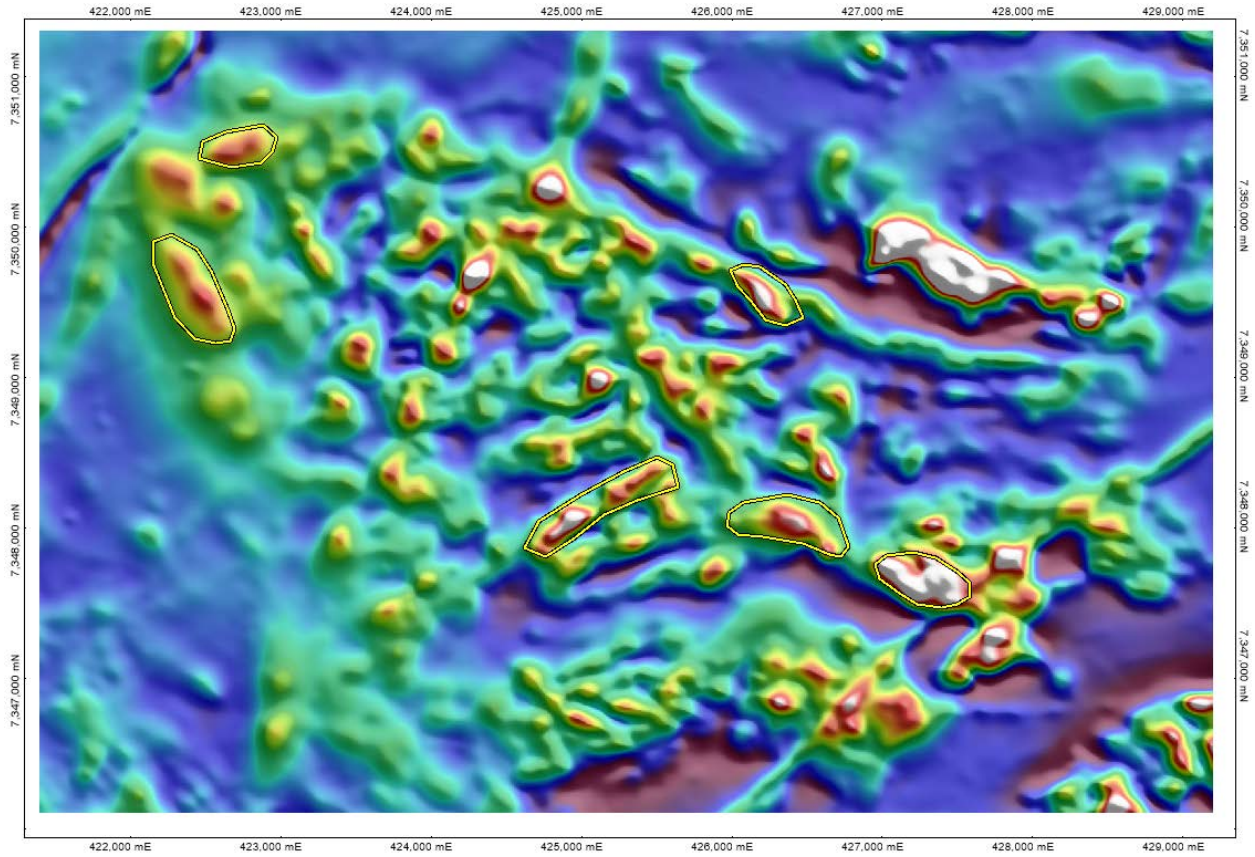


Figure 7. Features of interest highlighted from the depth filtering (yellow polygons) over RTP Upward-Continued 50m 1VD image (linear colour stretch).

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6 CONCLUSIONS & RECOMMENDATIONS

Forward and inversion modelling has been undertaken on magnetic data collected over Hastings' Yangibana project in 2016 in order to characterise prospects based on magnetic signature before follow-up work is performed.

It is believed that Target 1 represents a priority target worthy of more detailed investigation given the 2D/3D modelling that has been completed, this local magnetic body has a differing/stronger signature to other magnetic units in the local vicinity and is bounded by a major ~WNW-ESE structure along it's southern margin. It is believed that this magnetic unit is under thin cover. Optimised drill testing of both limbs of this Target 1 magnetic unit is recommended.

Targets 3 and 4 represent steeply dipping, relatively thin magnetic units of a similar nature to the Auer/Auer North prospects, and drill testing of both targets is recommended.

Target 6 appears to be representative of a shallow, magnetic paleochannel and follow up work such as shallow drill testing is recommended.

Both forward and inversion modelling has been performed at the Fraser Fault Zone prospect due to the complexity of the magnetic signal. These models should be integrated with the existing drilling and geological models in increase understanding and generate new drill targets.

The Auer forward modelling performed in 2016 was reviewed against new drill results and a mineralisation outline. The model is slightly shallower dipping and slightly thicker, but otherwise provides a good estimation of the behaviour of magnetic mineralisation horizon and is a successful proof-of-concept for future modelling studies in this project.

Depth filtering for the broader Target 5 area has defined a number of deeper/broader magnetic anomaly centres which should be assessed in more detail from a geological perspective/spatial context given the mapped structure/geology. Local 3D inversion of priority deeper anomalism deemed to be of geological interest at Target 5 is recommended to ascertain potential source depths/geometries.

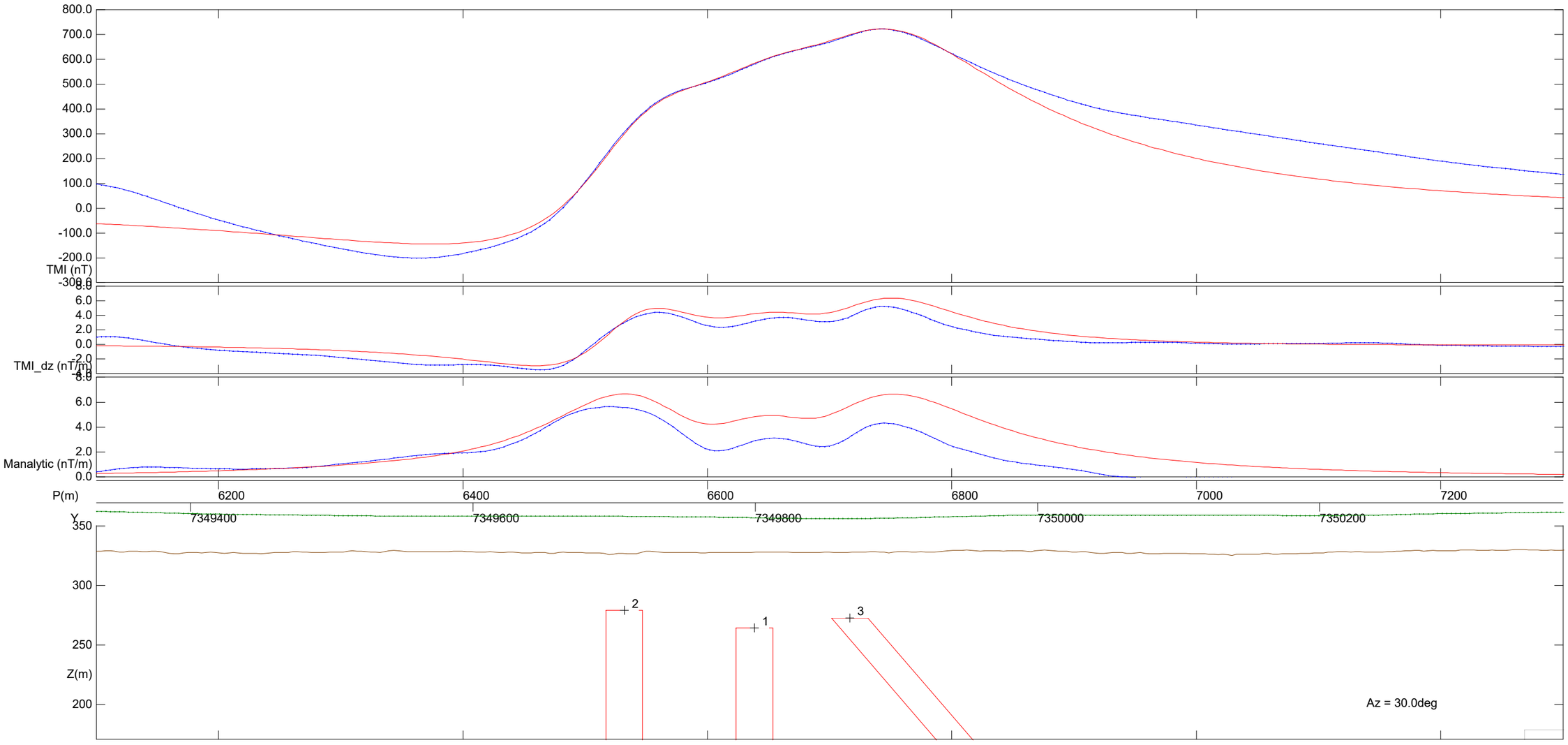
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APPENDIX A – POTENT MODELLING RESULTS

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TARGET 1

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Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

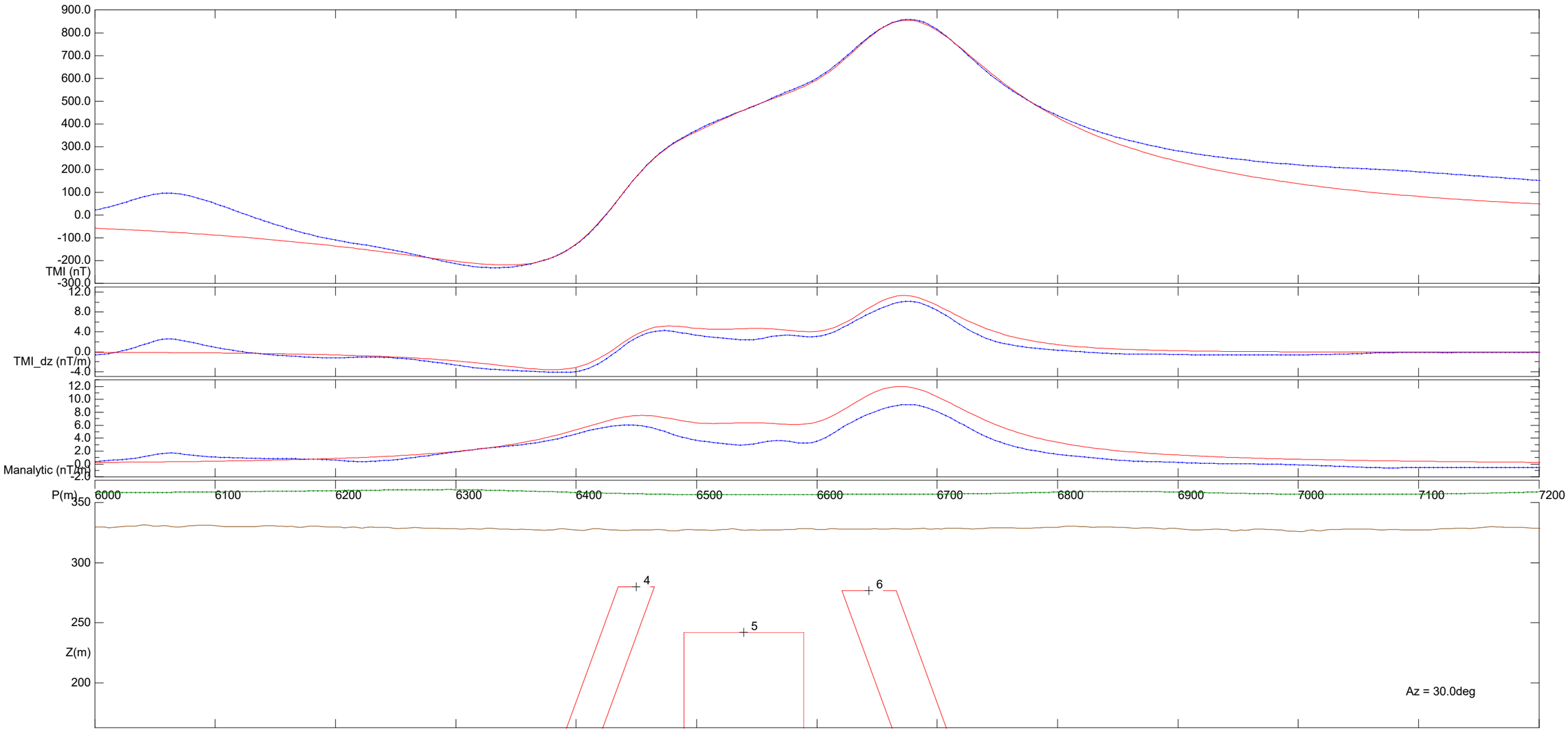
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2	Dyke	427327.4	7349707.1	278.9	-60.0	0.0	0.0	0.2000	30.0	120.0	400.0	90.0
3	Dyke	427419.7	7349866.9	272.4	-60.0	0.0	0.0	0.2000	30.0	120.0	400.0	50.0
4	Dyke	427443.5	7349608.4	280.0	-60.0	0.0	0.0	0.2500	30.0	120.0	400.0	110.0
5	Dyke	427488.3	7349685.9	242.0	-60.0	0.0	0.0	0.2500	100.0	120.0	400.0	90.0
6	Dyke	427541.1	7349775.5	276.9	-60.0	0.0	0.0	0.2500	45.0	120.0	400.0	70.0
9	Dyke	427792.0	7349661.8	282.5	-60.0	0.0	0.0	0.2000	50.0	120.0	400.0	60.0
10	Dyke	427715.4	7349529.2	281.5	-60.0	0.0	0.0	0.2000	20.0	120.0	400.0	90.0
11	Dyke	427757.2	7349601.5	263.0	-60.0	0.0	0.0	0.2000	40.0	120.0	400.0	90.0

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HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 1 - Line 110370

Map Reference: GDA94 MGA Zone 50		
Author: JT	Original scale: 1:3500	Report No.:
Drawn by: JT	Date: 11 May 2018	Plan No.:



Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
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2	Dyke	427327.4	7349707.1	278.9	-60.0	0.0	0.0	0.2000	30.0	120.0	400.0	90.0
3	Dyke	427419.7	7349866.9	272.4	-60.0	0.0	0.0	0.2000	30.0	120.0	400.0	50.0
4	Dyke	427443.5	7349608.4	280.0	-60.0	0.0	0.0	0.2500	30.0	120.0	400.0	110.0
5	Dyke	427488.3	7349685.9	242.0	-60.0	0.0	0.0	0.2500	100.0	120.0	400.0	90.0
6	Dyke	427541.1	7349775.5	276.9	-60.0	0.0	0.0	0.2500	45.0	120.0	400.0	70.0
9	Dyke	427792.0	7349661.8	282.5	-60.0	0.0	0.0	0.2000	50.0	120.0	400.0	60.0
10	Dyke	427715.4	7349529.2	281.5	-60.0	0.0	0.0	0.2000	20.0	120.0	400.0	90.0
11	Dyke	427757.2	7349601.5	263.0	-60.0	0.0	0.0	0.2000	40.0	120.0	400.0	90.0

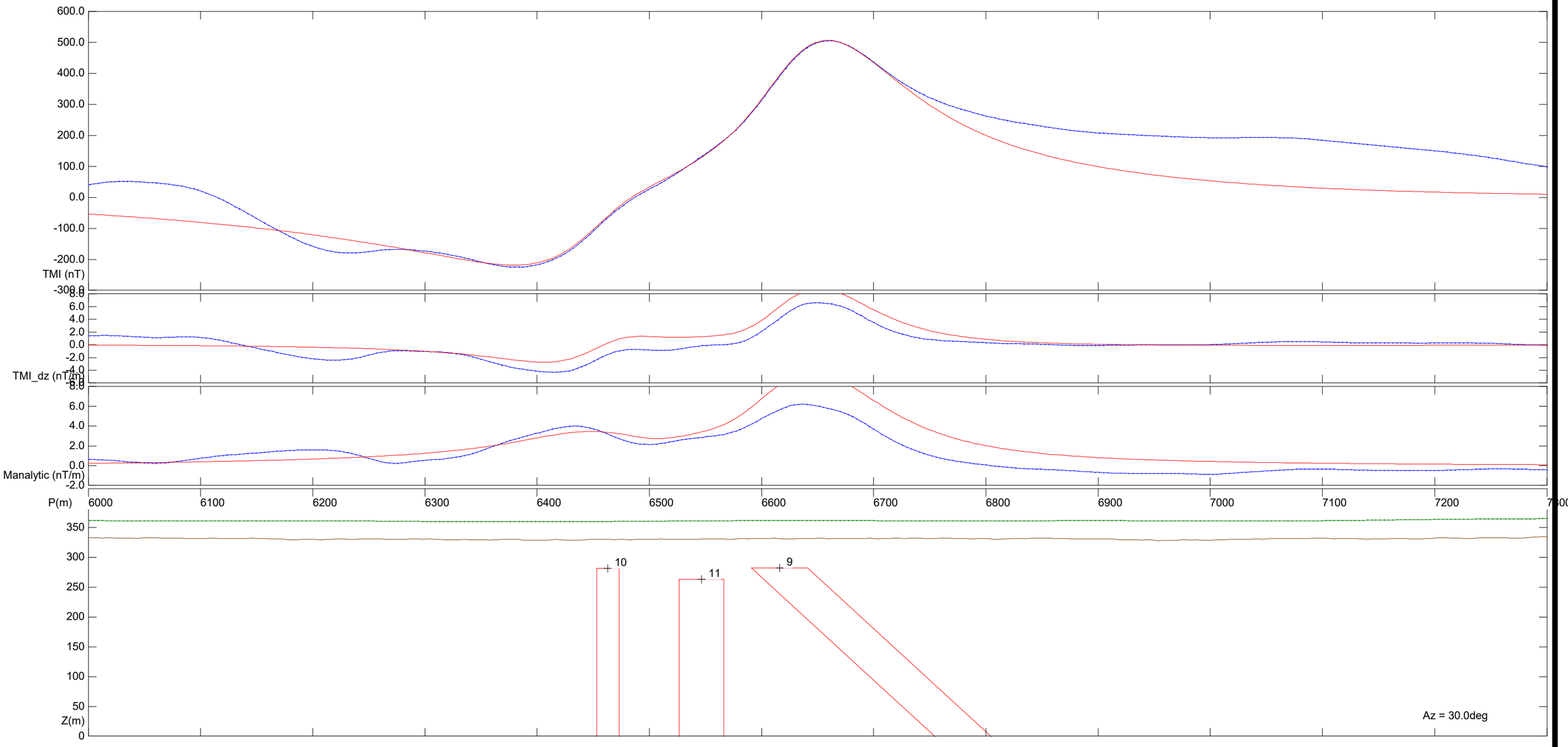
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HASTINGS TECHNOLOGY METALS YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 1 - Line 110430

Map Reference: GDA94 MGA Zone 50

Author: JT	Original scale: 1:3500	Report No.:
Drawn by: JT	Date: 11 May 2018	Plan No.:



Model Summary

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2	Dyke	427327.4	7349707.1	278.9	-60.0	0.0	0.0	0.2000	30.0	120.0	400.0	90.0
3	Dyke	427419.7	7349866.9	272.4	-60.0	0.0	0.0	0.2000	30.0	120.0	400.0	50.0
4	Dyke	427443.5	7349608.4	280.0	-60.0	0.0	0.0	0.2500	30.0	120.0	400.0	110.0
5	Dyke	427488.3	7349685.9	242.0	-60.0	0.0	0.0	0.2500	100.0	120.0	400.0	90.0
6	Dyke	427541.1	7349775.5	276.9	-60.0	0.0	0.0	0.2500	45.0	120.0	400.0	70.0
9	Dyke	427792.0	7349661.8	282.5	-60.0	0.0	0.0	0.2000	50.0	120.0	400.0	60.0
10	Dyke	427715.4	7349529.2	281.5	-60.0	0.0	0.0	0.2000	20.0	120.0	400.0	90.0
11	Dyke	427757.2	7349601.5	263.0	-60.0	0.0	0.0	0.2000	40.0	120.0	400.0	90.0

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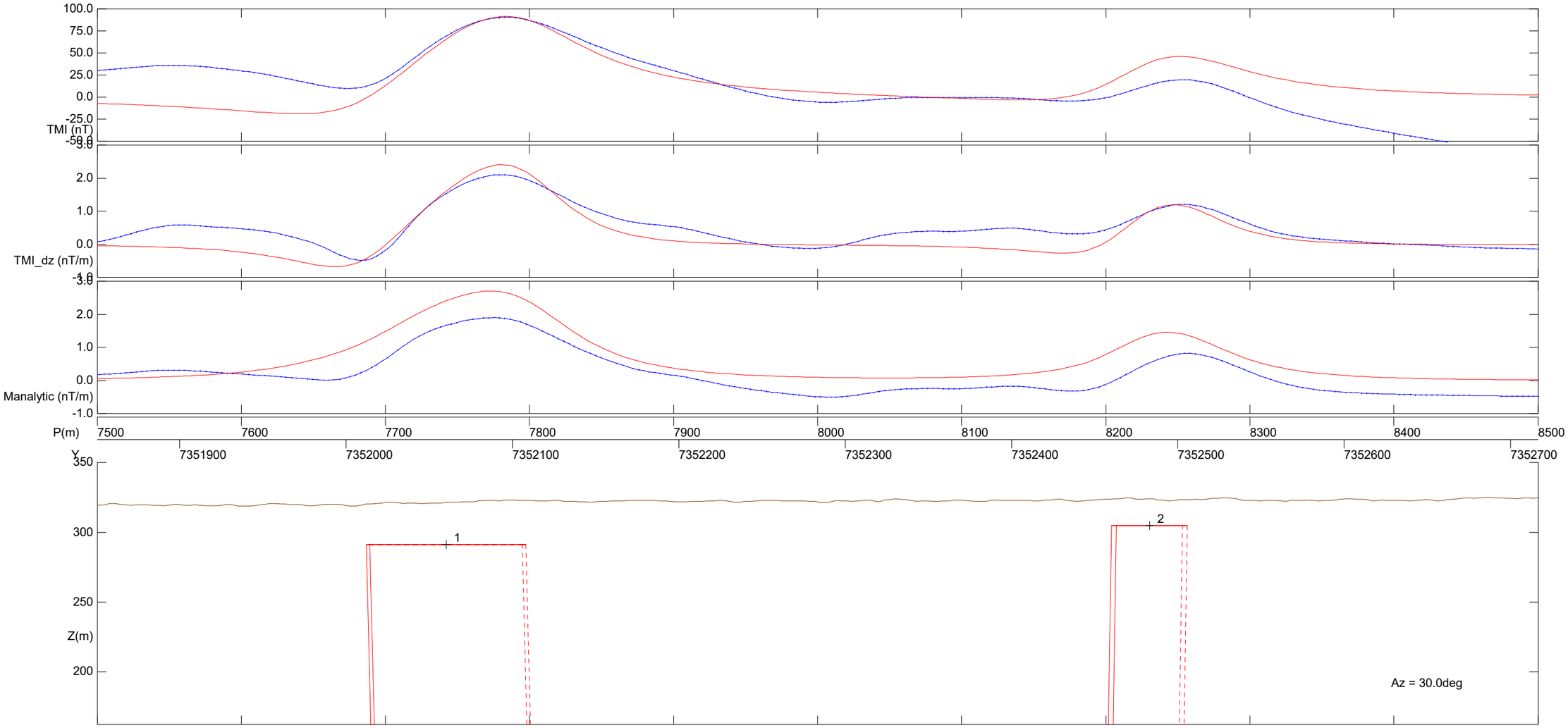
HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 1 - Line 110540

Map Reference: GDA94 MGA Zone 50		
Author: JT	Original scale: 1:3500	Report No.:
Drawn by: JT	Date: 11 May 2018	Plan No.:

TARGET 3

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Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
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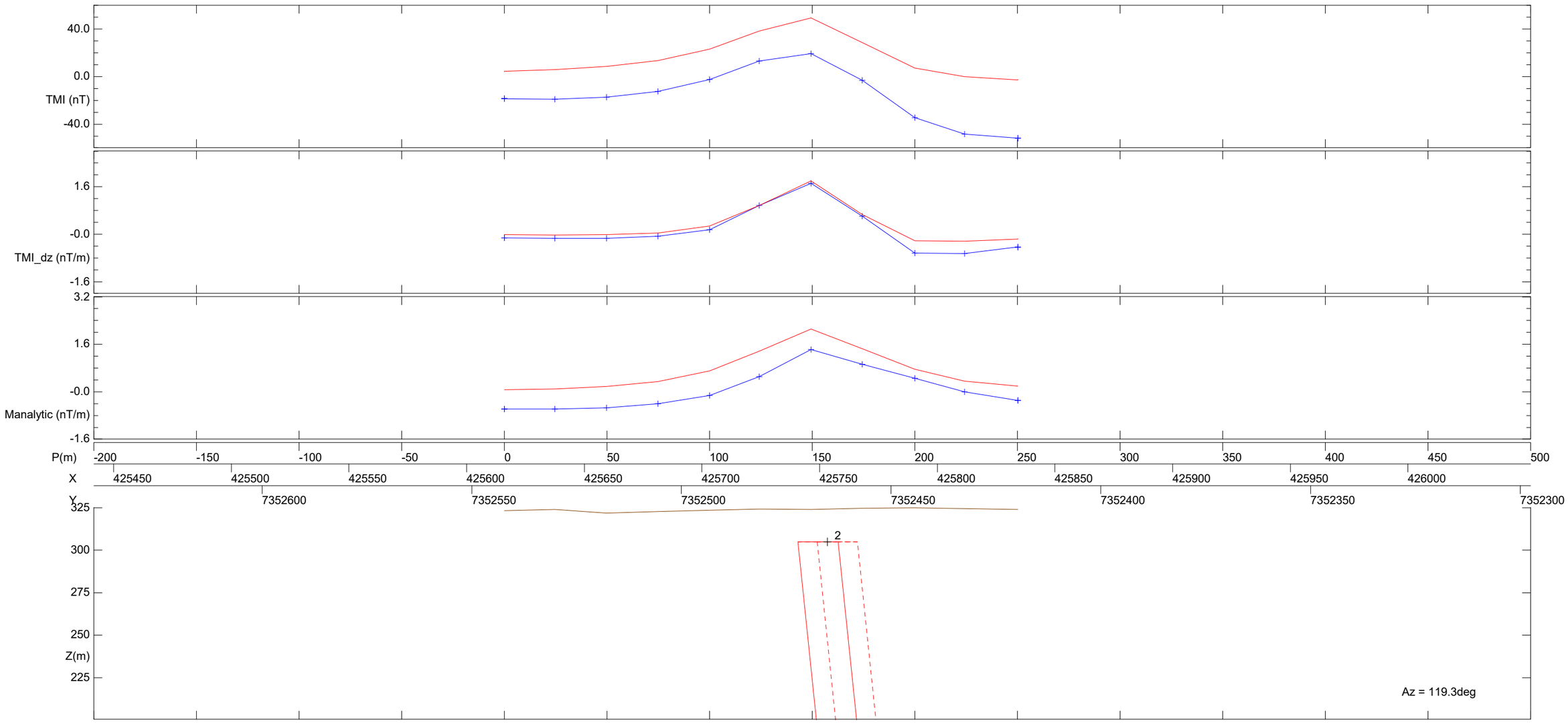
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HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 3 - Line 109260

Map Reference: GDA94 MGA Zone 50

Author: JT	Original scale: 1:3000	Report No.:
Drawn by: JT	Date: 11 May 2018	Plan No.:



Model Summary

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1	Dyke	425478.8	7352061.8	291.1	40.0	0.0	0.0	0.1000	15.0	110.0	400.0	98.0
2	Dyke	425754.1	7352466.5	304.8	40.0	0.0	0.0	0.0500	20.0	50.0	400.0	85.0

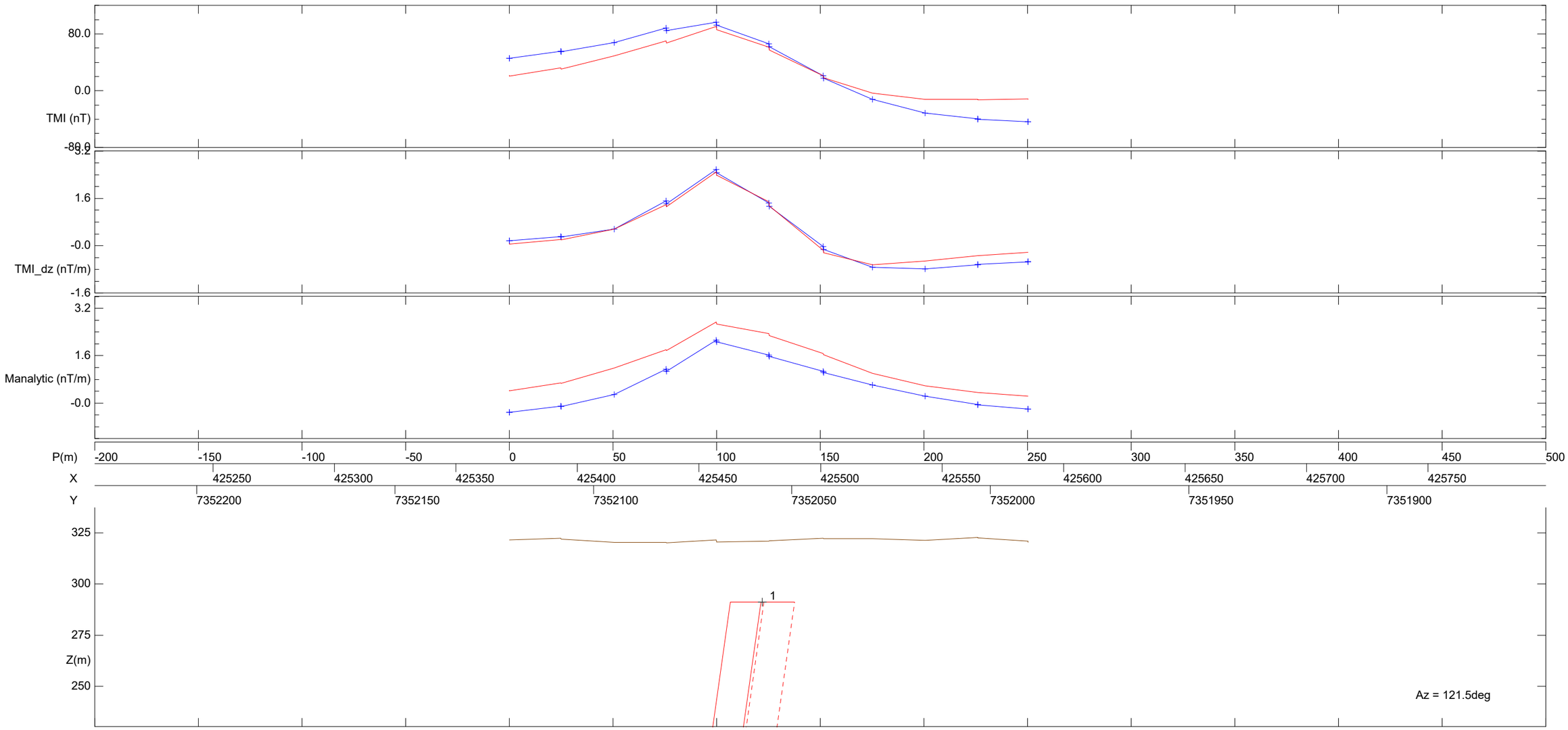
Southern Geoscience Consultants

HASTINGS TECHNOLOGY METALS YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 3 - North Anomaly X-Section

Map Reference: GDA94 MGA Zone 50

Author: JT	Original scale: 1:2000	Report No.:
Drawn by: JT	Date: 11 May 2018	Plan No.:



Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	425478.8	7352061.8	291.1	40.0	0.0	0.0	0.1000	15.0	110.0	400.0	98.0
2	Dyke	425754.1	7352466.5	304.8	40.0	0.0	0.0	0.0500	20.0	50.0	400.0	85.0

Southern Geoscience Consultants

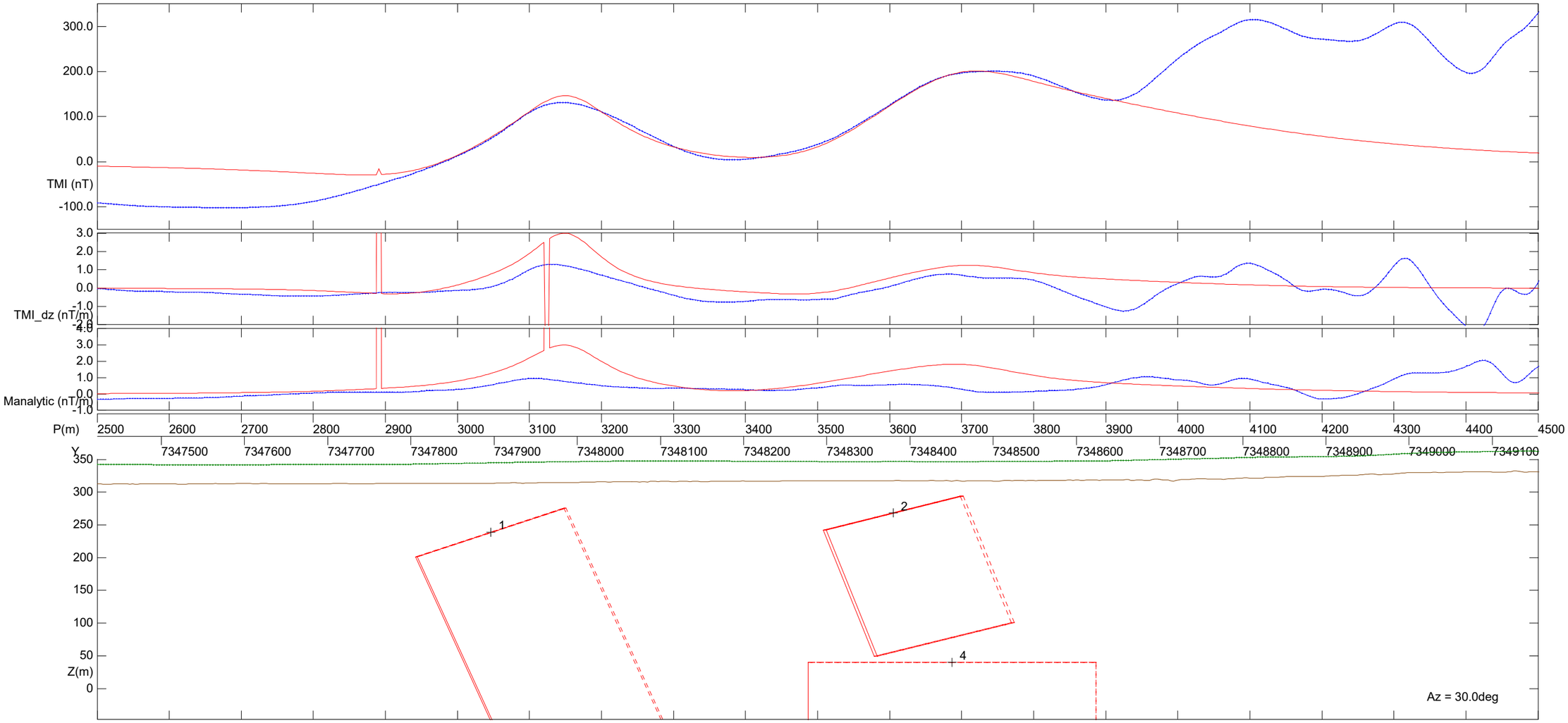
HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 3 - South Anomaly X-Section

Map Reference: GDA94 MGA Zone 50		
Author: JT	Original scale: 1:2000	Report No.:
Drawn by: JT	Date: 11 May 2018	Plan No.:

TARGET 4

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Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	423366.0	7347896.0	238.4	25.0	0.0	-20.0	0.1000	30.0	220.0	500.0	60.0
2	Dyke	423733.3	7348329.0	268.2	40.0	0.0	-15.0	0.1000	23.0	200.0	200.0	120.0
3	Dyke	423593.8	7347840.6	218.4	90.0	0.0	0.0	0.1000	15.0	250.0	200.0	90.0
4	Dyke	423908.6	7348322.0	39.8	30.0	0.0	0.0	0.0500	400.0	400.0	400.0	90.0
5	Dyke	423826.6	7348244.1	230.0	-50.0	0.0	0.0	0.1000	55.0	70.0	150.0	90.0
6	Dyke	423938.5	7348241.2	254.2	-50.0	0.0	0.0	0.1000	20.0	160.0	100.0	90.0
7	Dyke	423241.6	7348044.5	210.6	30.0	0.0	0.0	-0.1000	60.0	100.0	200.0	90.0

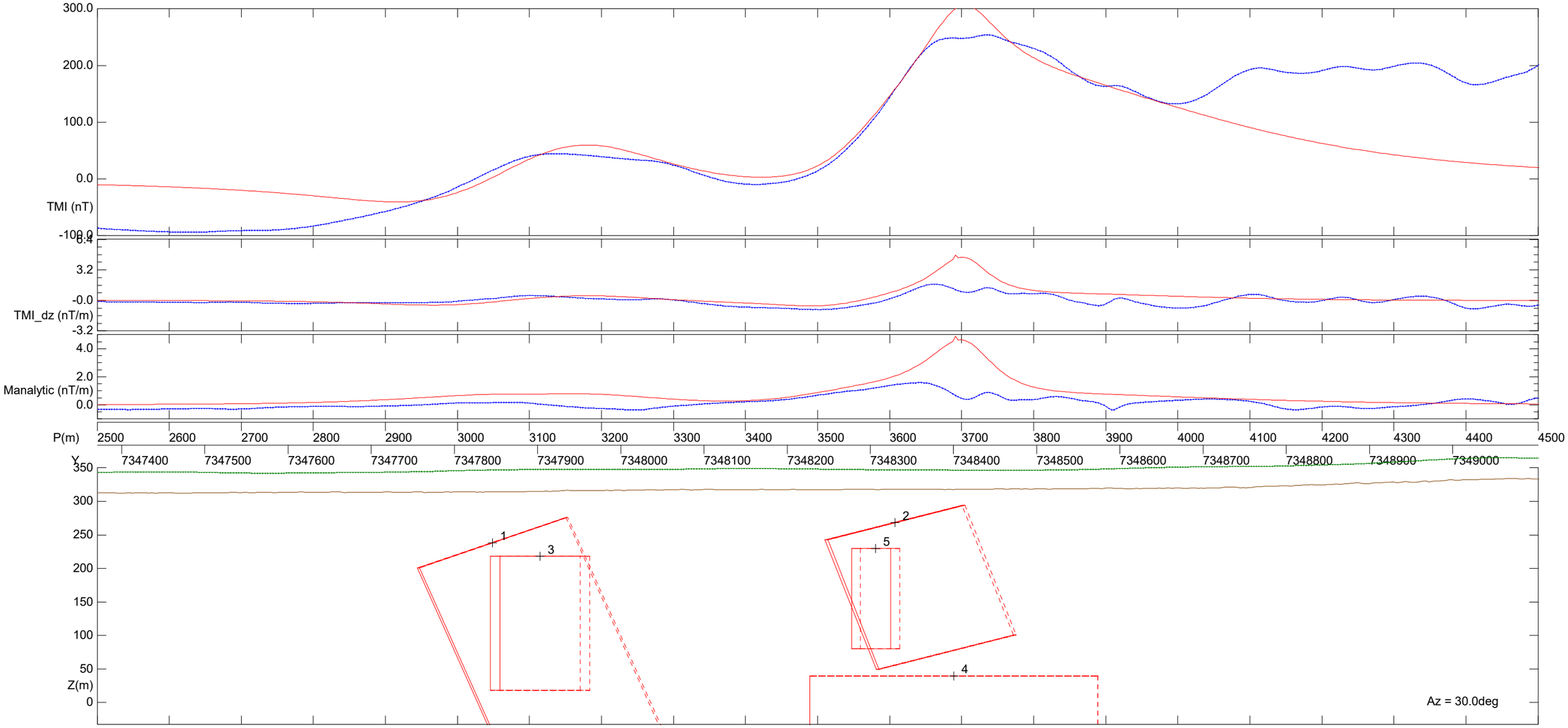
Southern Geoscience Consultants

HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 4 - Line 109360

Map Reference: GDA94 MGA Zone 50

Author: JT	Original scale: 1:6000	Report No.:
Drawn by: JT	Date: 14 May 2018	Plan No.:



Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

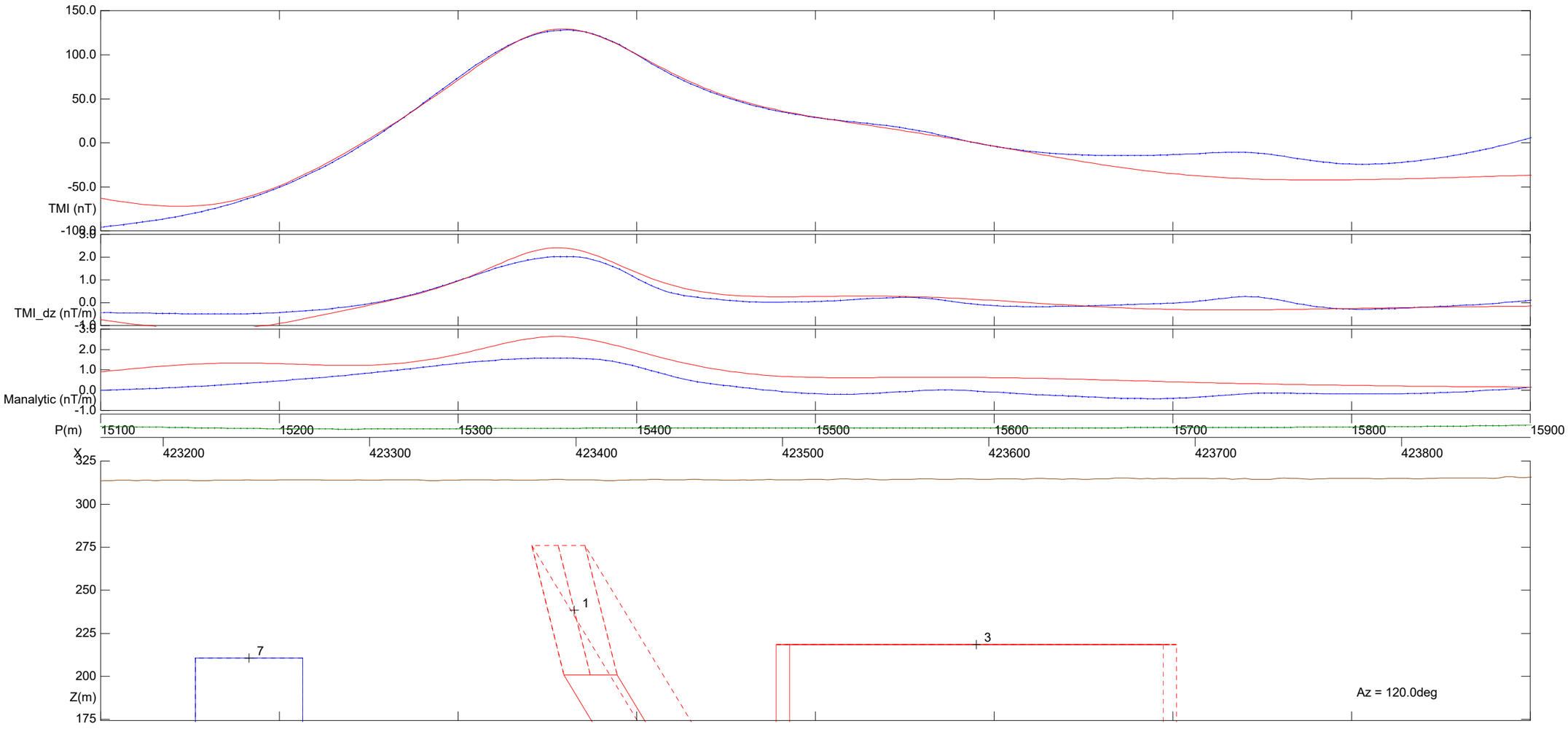
Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	423366.0	7347896.0	238.4	25.0	0.0	-20.0	0.1000	30.0	220.0	500.0	60.0
2	Dyke	423733.3	7348329.0	268.2	40.0	0.0	-15.0	0.1000	23.0	200.0	200.0	120.0
3	Dyke	423593.8	7347840.6	218.4	90.0	0.0	0.0	0.1000	15.0	250.0	200.0	90.0
4	Dyke	423908.6	7348322.0	39.8	30.0	0.0	0.0	0.0500	400.0	400.0	400.0	90.0
5	Dyke	423826.6	7348244.1	230.0	-50.0	0.0	0.0	0.1000	55.0	70.0	150.0	90.0
6	Dyke	423938.5	7348241.2	254.2	-50.0	0.0	0.0	0.1000	20.0	160.0	100.0	90.0
7	Dyke	423241.6	7348044.5	210.6	30.0	0.0	0.0	-0.1000	60.0	100.0	200.0	90.0

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HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 4 - Line 109400

Map Reference: GDA94 MGA Zone 50		
Author: JT	Original scale: 1:6000	Report No.:
Drawn by: JT	Date: 14 May 2018	Plan No.:



Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

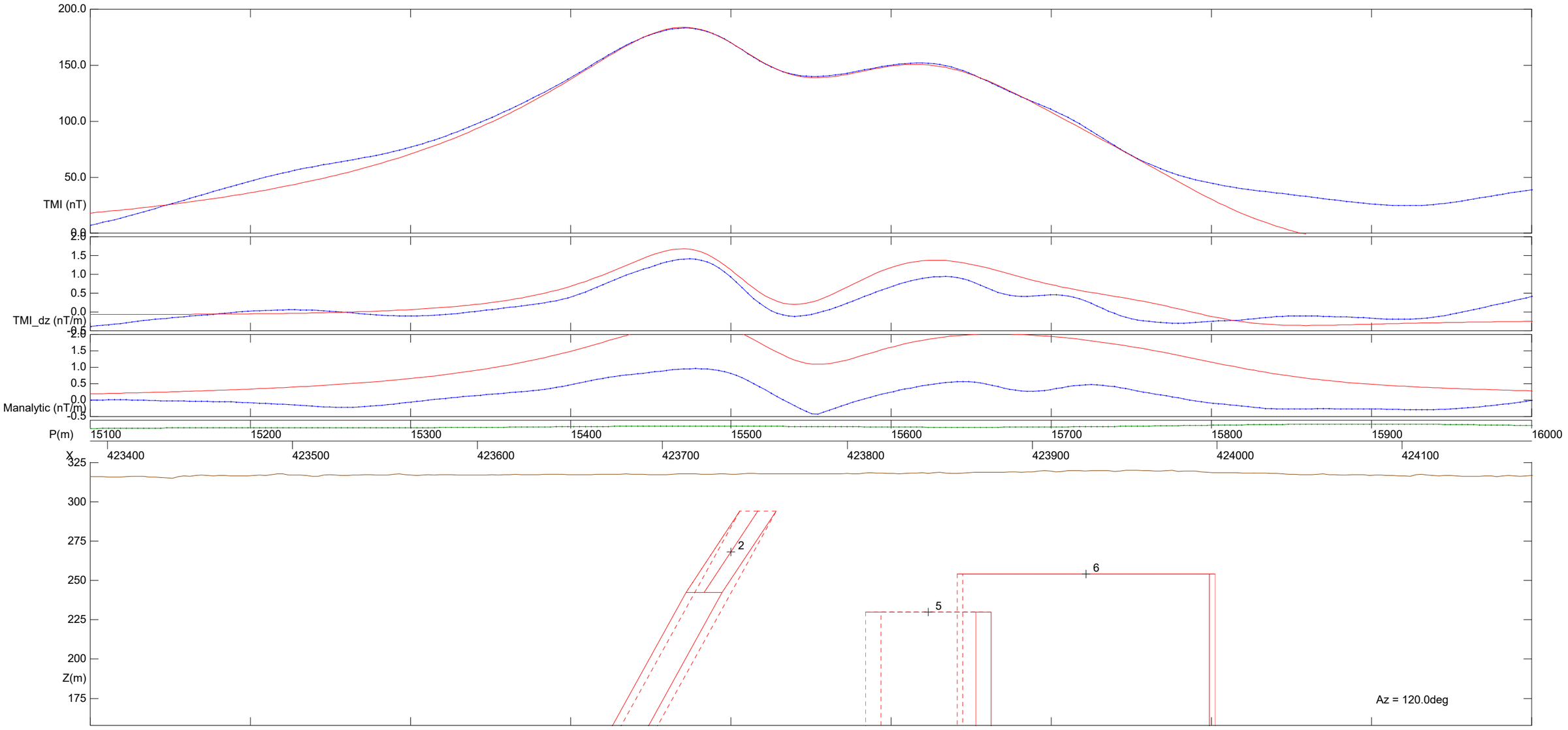
Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	423366.0	7347896.0	238.4	25.0	0.0	-20.0	0.1000	30.0	220.0	500.0	60.0
2	Dyke	423733.3	7348329.0	268.2	40.0	0.0	-15.0	0.1000	23.0	200.0	200.0	120.0
3	Dyke	423593.8	7347840.6	218.4	90.0	0.0	0.0	0.1000	15.0	250.0	200.0	90.0
4	Dyke	423908.6	7348322.0	39.8	30.0	0.0	0.0	0.0500	400.0	400.0	400.0	90.0
5	Dyke	423826.6	7348244.1	230.0	-50.0	0.0	0.0	0.1000	55.0	70.0	150.0	90.0
6	Dyke	423938.5	7348241.2	254.2	-50.0	0.0	0.0	0.1000	20.0	160.0	100.0	90.0
7	Dyke	423241.6	7348044.5	210.6	30.0	0.0	0.0	-0.1000	60.0	100.0	200.0	90.0

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HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 4 - Line 190170

Map Reference: GDA94 MGA Zone 50		
Author: JT	Original scale: 1:2500	Report No.:
Drawn by: JT	Date: 14 May 2018	Plan No.:



Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	423366.0	7347896.0	238.4	25.0	0.0	-20.0	0.1000	30.0	220.0	500.0	60.0
2	Dyke	423733.3	7348329.0	268.2	40.0	0.0	-15.0	0.1000	23.0	200.0	200.0	120.0
3	Dyke	423593.8	7347840.6	218.4	90.0	0.0	0.0	0.1000	15.0	250.0	200.0	90.0
4	Dyke	423908.6	7348322.0	39.8	30.0	0.0	0.0	0.0500	400.0	400.0	400.0	90.0
5	Dyke	423826.6	7348244.1	230.0	-50.0	0.0	0.0	0.1000	55.0	70.0	150.0	90.0
6	Dyke	423938.5	7348241.2	254.2	-50.0	0.0	0.0	0.1000	20.0	160.0	100.0	90.0
7	Dyke	423241.6	7348044.5	210.6	30.0	0.0	0.0	-0.1000	60.0	100.0	200.0	90.0

Southern Geoscience Consultants

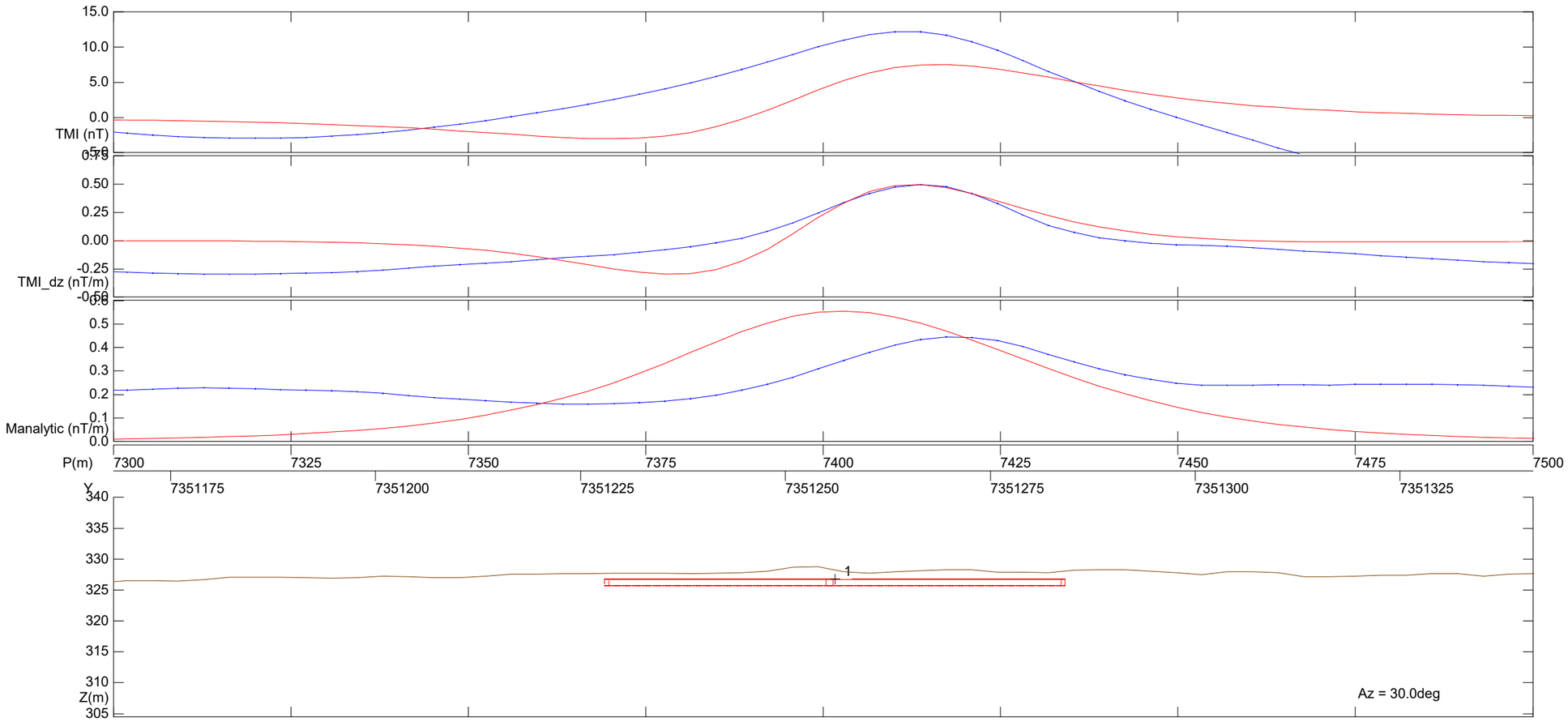
HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
TARGET 4 - Line 190190

Map Reference: GDA94 MGA Zone 50		
Author: JT	Original scale: 1:2500	Report No.:
Drawn by: JT	Date: 14 May 2018	Plan No.:

TARGET 6

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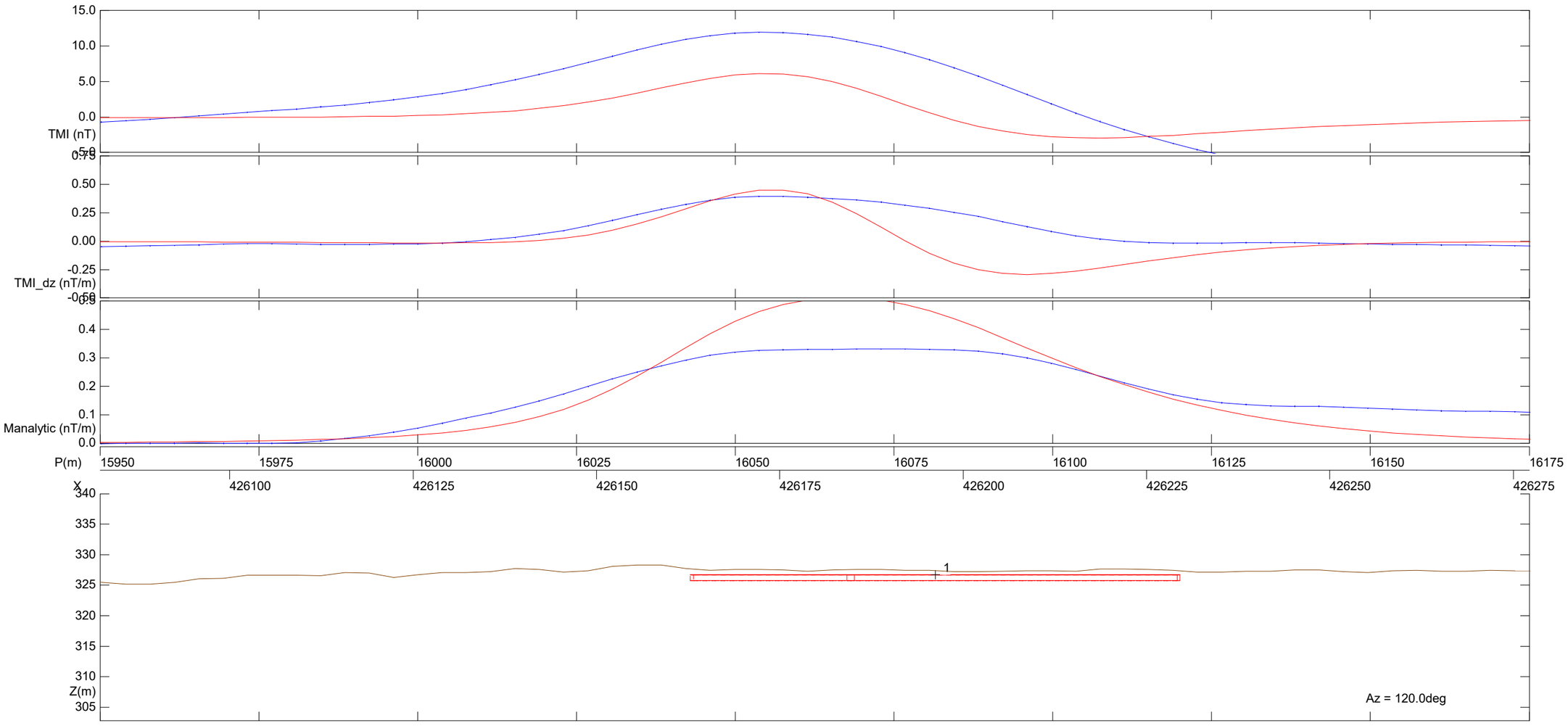


Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	426201.8	7351255.6	326.7	-100.0	0.0	0.0	1.3000	0.8	100.0	1.0	90.0

Southern Geoscience Consultants		
HASTINGS TECHNOLOGY METALS YANGIBANA PROJECT		
Airborne Magnetic Forward Modelling TARGET 6 - Line 109670		
Map Reference: GDA94 MGA Zone 50		
Author: JT	Original scale: 1:700	Report No.:
Drawn by: JT	Date: 29 May 2018	Plan No.:



Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	426201.8	7351255.6	326.7	-100.0	0.0	0.0	1.3000	0.8	100.01.0	90.0	

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HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

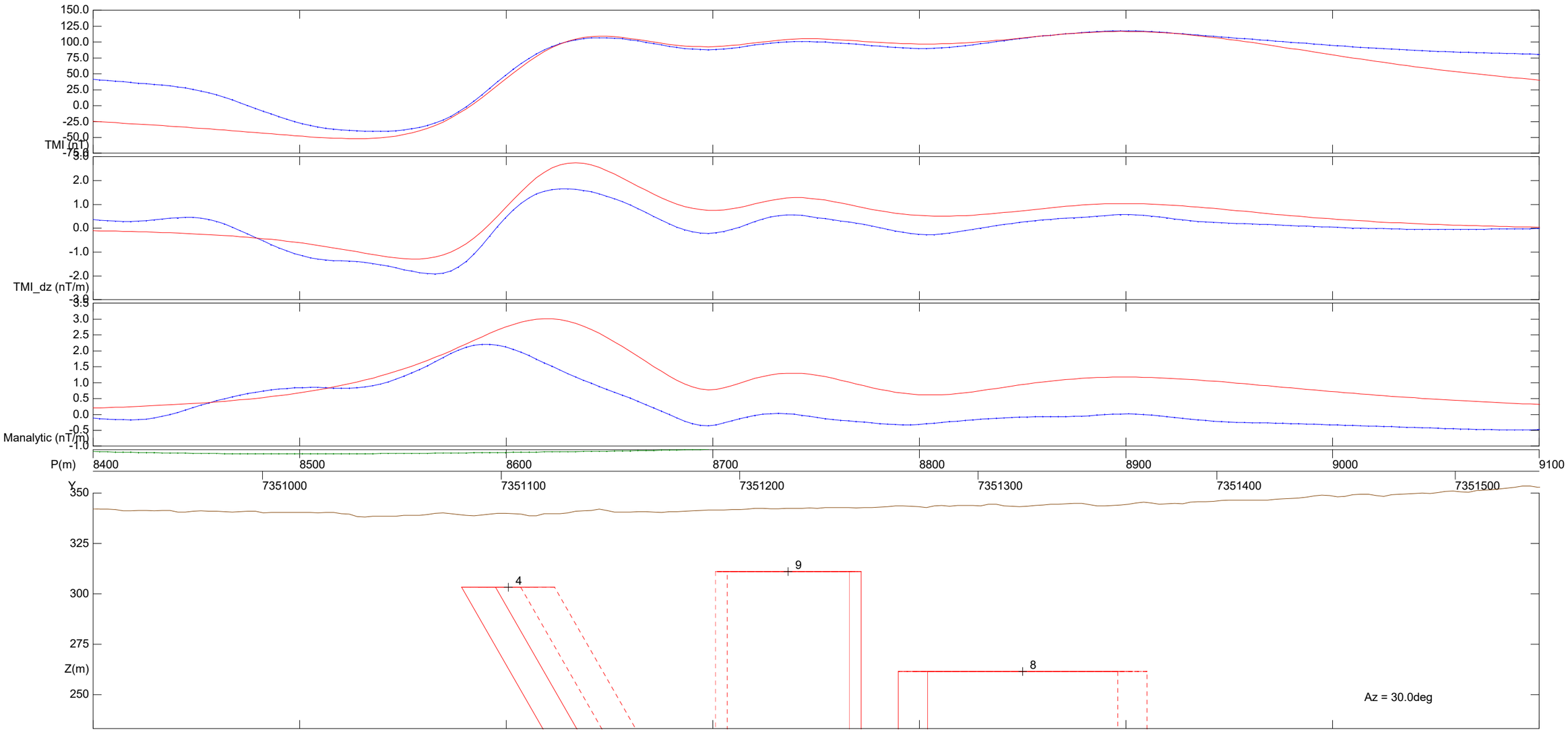
Airborne Magnetic Forward Modelling
TARGET 6 - Line 190340

Map Reference: GDA94 MGA Zone 50

Author: JT	Original scale: 1:700	Report No.:
Drawn by: JT	Date: 29 May 2018	Plan No.:

FRASER FAULT ZONE

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Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	429508.2	7351034.1	305.9	-80.0	0.0	0.0	0.1000	11.0	250.0	200.0	90.0
2	Dyke	429784.1	7350962.1	338.4	-102.0	0.0	0.0	0.1000	7.0	90.0	100.0	90.0
3	Dyke	429823.1	7351013.5	341.0	-10.0	0.0	0.0	0.1000	4.0	80.0	100.0	90.0
4	Dyke	429334.2	7351092.9	303.4	-45.0	0.0	0.0	0.1500	17.0	110.0	200.0	60.0
5	Dyke	429821.6	7351172.7	274.0	30.0	0.0	0.0	0.0500	70.0	150.0	400.0	90.0
6	Dyke	429566.1	7351291.3	319.3	-10.0	0.0	0.0	0.1000	25.0	25.0	40.0	90.0
7	Dyke	430125.5	7350997.2	285.2	10.0	0.0	0.0	0.0500	30.0	150.0	400.0	90.0
8	Dyke	429474.5	7351299.5	261.5	-15.0	0.0	0.0	0.1000	20.0	150.0	200.0	90.0
9	Dyke	429422.7	7351198.4	311.1	-75.0	0.0	0.0	0.1000	6.0	250.0	200.0	90.0
11	Dyke	429566.8	7351135.6	278.5	-80.0	0.0	0.0	0.0500	100.0	100.0	20.0	90.0
13	Dyke	429595.3	7351274.4	223.6	-10.0	0.0	0.0	0.1000	100.0	100.0	100.0	90.0
14	Dyke	429848.1	7351056.8	335.8	-10.0	0.0	0.0	0.1000	2.0	80.0	100.0	90.0

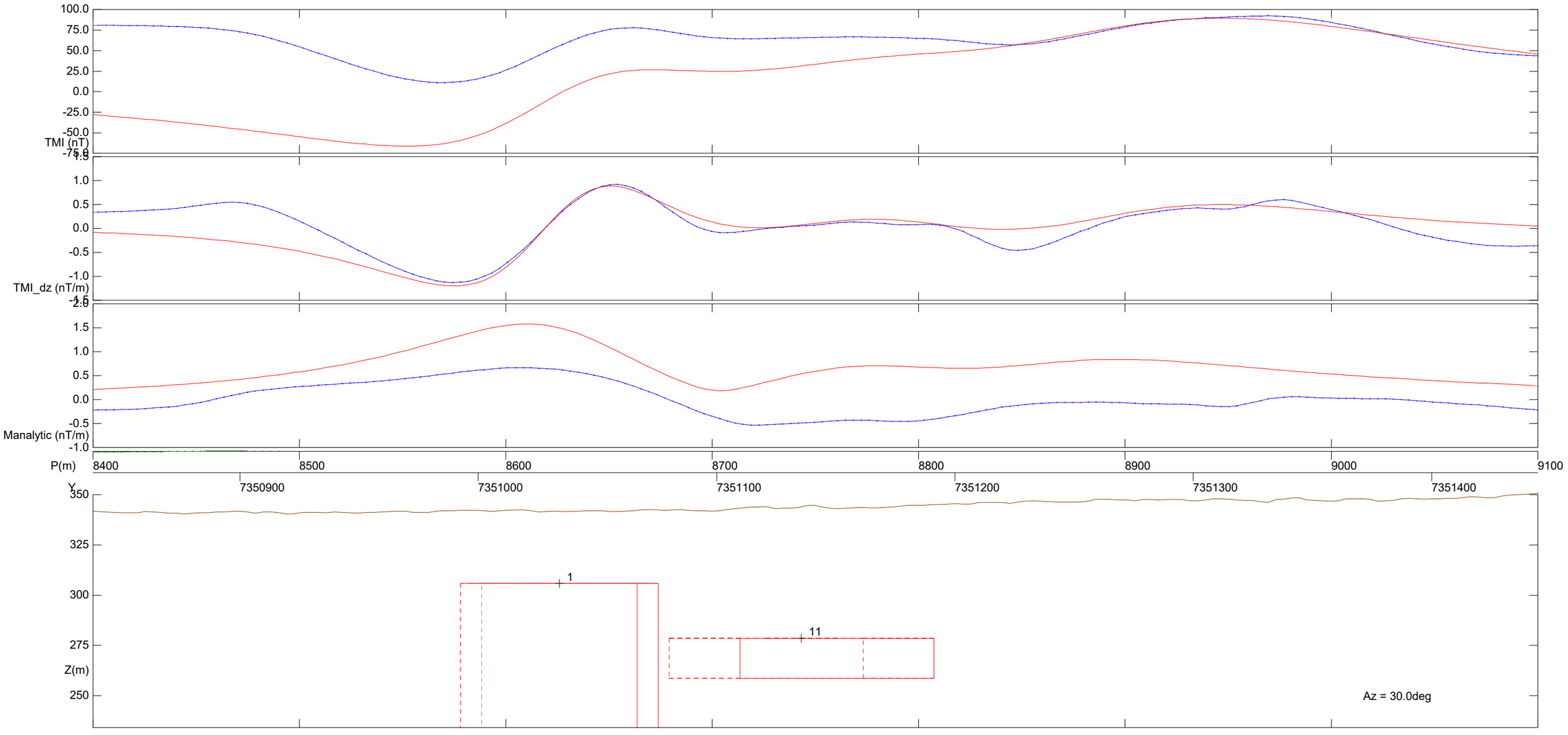
Southern Geoscience Consultants

HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
FRASER FAULT ZONE - Line 110780

Map Reference: GDA94 MGA Zone 50

Author: JT	Original scale: 1:2000	Report No.:
Drawn by: JT	Date: 29 May 2018	Plan No.:



Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	429508.2	7351034.1	305.9	-80.0	0.0	0.0	0.1000	11.0	250.0	200.0	90.0
2	Dyke	429784.1	7350962.1	338.4	-102.0	0.0	0.0	0.1000	7.0	90.0	100.0	90.0
3	Dyke	429823.1	7351013.5	341.0	-10.0	0.0	0.0	0.1000	4.0	80.0	100.0	90.0
4	Dyke	429334.2	7351092.9	303.4	-45.0	0.0	0.0	0.1500	17.0	110.0	200.0	60.0
5	Dyke	429821.6	7351172.7	274.0	30.0	0.0	0.0	0.0500	70.0	150.0	400.0	90.0
6	Dyke	429566.1	7351291.3	319.3	-10.0	0.0	0.0	0.1000	25.0	25.0	40.0	90.0
7	Dyke	430125.5	7350997.2	285.2	10.0	0.0	0.0	0.0500	30.0	150.0	400.0	90.0
8	Dyke	429474.5	7351299.5	261.5	-15.0	0.0	0.0	0.1000	20.0	150.0	200.0	90.0
9	Dyke	429422.7	7351198.4	311.1	-75.0	0.0	0.0	0.1000	6.0	250.0	200.0	90.0
11	Dyke	429566.8	7351135.6	278.5	-80.0	0.0	0.0	0.0500	100.0	100.0	20.0	90.0
13	Dyke	429595.3	7351274.4	223.6	-10.0	0.0	0.0	0.1000	100.0	100.0	100.0	90.0
14	Dyke	429848.1	7351056.8	335.8	-10.0	0.0	0.0	0.1000	2.0	80.0	100.0	90.0

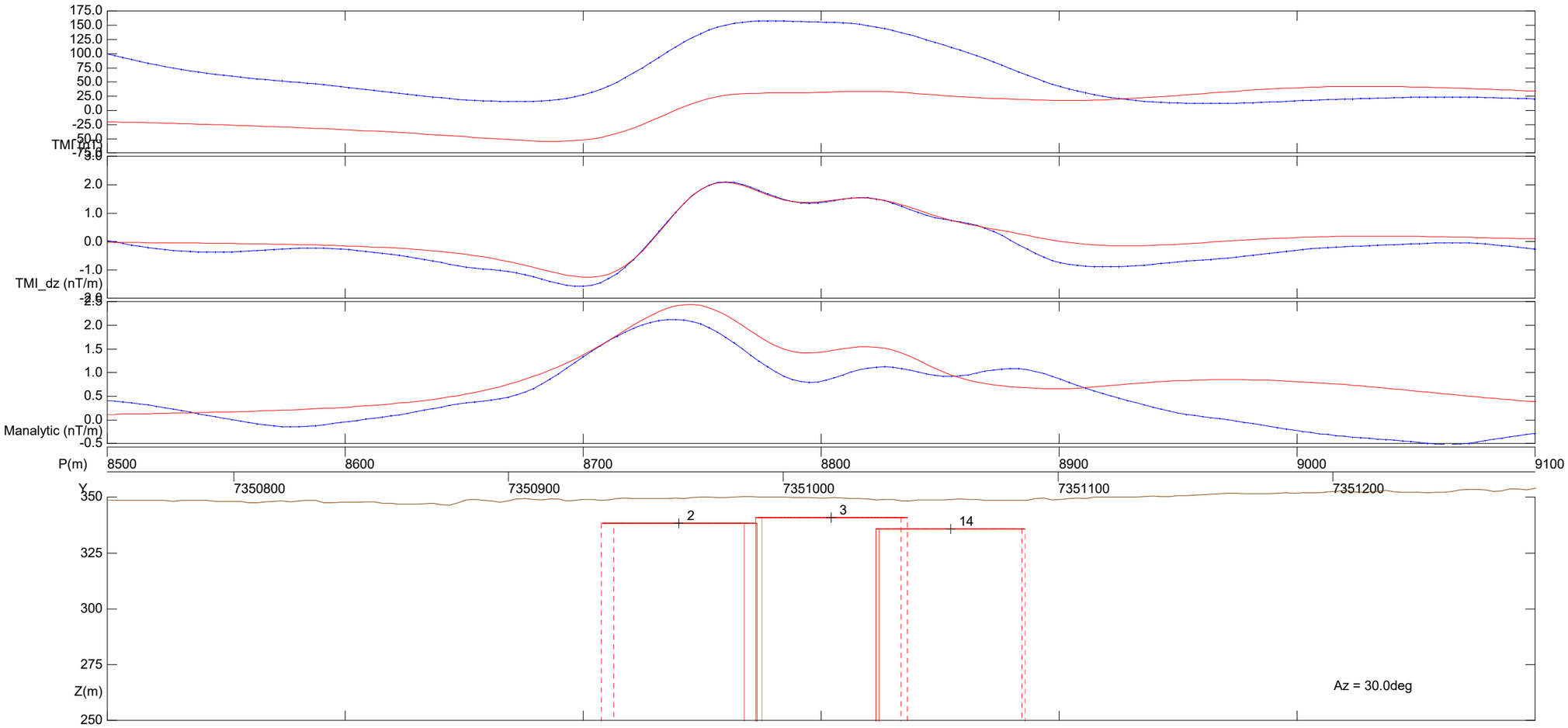
Southern Geoscience Consultants

HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
FRASER FAULT ZONE - Line 110860

Map Reference: GDA94 MGA Zone 50

Author: JT	Original scale: 1:2000	Report No.:
Drawn by: JT	Date: 29 May 2018	Plan No.:



Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	429508.2	7351034.1	305.9	-80.0	0.0	0.0	0.1000	11.0	250.0	200.0	90.0
2	Dyke	429784.1	7350962.1	338.4	-102.0	0.0	0.0	0.1000	7.0	90.0	100.0	90.0
3	Dyke	429823.1	7351013.5	341.0	-10.0	0.0	0.0	0.1000	4.0	80.0	100.0	90.0
4	Dyke	429334.2	7351092.9	303.4	-45.0	0.0	0.0	0.1500	17.0	110.0	200.0	60.0
5	Dyke	429821.6	7351172.7	274.0	30.0	0.0	0.0	0.0500	70.0	150.0	400.0	90.0
6	Dyke	429566.1	7351291.3	319.3	-10.0	0.0	0.0	0.1000	25.0	25.0	40.0	90.0
7	Dyke	430125.5	7350997.2	285.2	10.0	0.0	0.0	0.0500	30.0	150.0	400.0	90.0
8	Dyke	429474.5	7351299.5	261.5	-15.0	0.0	0.0	0.1000	20.0	150.0	200.0	90.0
9	Dyke	429422.7	7351198.4	311.1	-75.0	0.0	0.0	0.1000	6.0	250.0	200.0	90.0
11	Dyke	429566.8	7351135.6	278.5	-80.0	0.0	0.0	0.0500	100.0	100.0	20.0	90.0
13	Dyke	429595.3	7351274.4	223.6	-10.0	0.0	0.0	0.1000	100.0	100.0	100.0	90.0
14	Dyke	429848.1	7351056.8	335.8	-10.0	0.0	0.0	0.1000	2.0	80.0	100.0	90.0

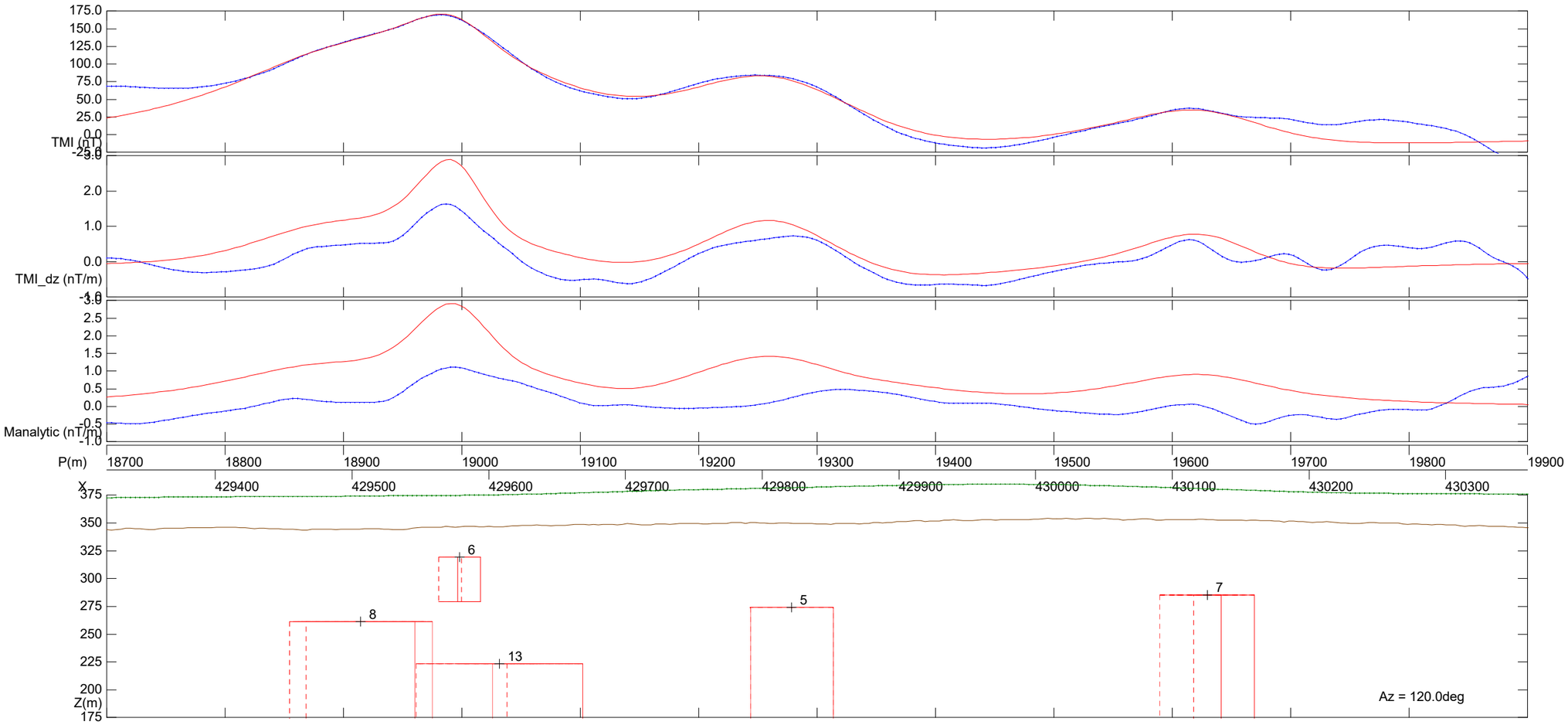
Southern Geoscience Consultants

HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
FRASER FAULT ZONE - Line 110970

Map Reference: GDA94 MGA Zone 50

Author: JT	Original scale: 1:2000	Report No.:
Drawn by: JT	Date: 29 May 2018	Plan No.:



Model Summary

IGRF : H = 53901; Az = 0.7; Inc = -57.3

Body	Type	X	Y	Z	Strike	Dip	Plunge	Susc.	A	B	C	Slope
1	Dyke	429508.2	7351034.1	305.9	-80.0	0.0	0.0	0.1000	11.0	250.0	200.0	90.0
2	Dyke	429784.1	7350962.1	338.4	-102.0	0.0	0.0	0.1000	7.0	90.0	100.0	90.0
3	Dyke	429823.1	7351013.5	341.0	-10.0	0.0	0.0	0.1000	4.0	80.0	100.0	90.0
4	Dyke	429334.2	7351092.9	303.4	-45.0	0.0	0.0	0.1500	17.0	110.0	200.0	60.0
5	Dyke	429821.6	7351172.7	274.0	30.0	0.0	0.0	0.0500	70.0	150.0	400.0	90.0
6	Dyke	429566.1	7351291.3	319.3	-10.0	0.0	0.0	0.1000	25.0	25.0	40.0	90.0
7	Dyke	430125.5	7350997.2	285.2	10.0	0.0	0.0	0.0500	30.0	150.0	400.0	90.0
8	Dyke	429474.5	7351299.5	261.5	-15.0	0.0	0.0	0.1000	20.0	150.0	200.0	90.0
9	Dyke	429422.7	7351198.4	311.1	-75.0	0.0	0.0	0.1000	6.0	250.0	200.0	90.0
11	Dyke	429566.8	7351135.6	278.5	-80.0	0.0	0.0	0.0500	100.0	100.0	20.0	90.0
13	Dyke	429595.3	7351274.4	223.6	-10.0	0.0	0.0	0.1000	100.0	100.0	100.0	90.0
14	Dyke	429848.1	7351056.8	335.8	-10.0	0.0	0.0	0.1000	2.0	80.0	100.0	90.0

Southern Geoscience Consultants

HASTINGS TECHNOLOGY METALS
YANGIBANA PROJECT

Airborne Magnetic Forward Modelling
FRASER FAULT ZONE - Line 190430

Map Reference: GDA94 MGA Zone 50

Author: JT	Original scale: 1:4000	Report No.:
Drawn by: JT	Date: 29 May 2018	Plan No.: